

Midazolam: Drug information - UpToDate

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For additional information see "Midazolam: Patient drug information" and "Midazolam: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

ALERT: US Boxed Warning

Personnel and equipment for monitoring and resuscitation:

Prefilled syringes and vials:

Intravenous midazolam has been associated with respiratory depression and respiratory arrest, especially when used for sedation in noncritical care settings. In some cases, where this was not recognized promptly and treated effectively, death or hypoxic encephalopathy has resulted. Intravenous midazolam should be used only in hospital or ambulatory care settings, including physicians' and dental offices, that provide for continuous monitoring of respiratory and cardiac function (eg, pulse oximetry). Immediate availability of resuscitative drugs and age- and size-appropriate equipment for bag/valve/mask ventilation and intubation, and personnel trained in their use and skilled in airway management should be assured. For deeply sedated pediatric patients, a dedicated individual, other than the practitioner performing the procedure, should monitor the patient throughout the procedure.

Premixed IV solutions:

Only personnel trained in the administration of procedural sedation, and not involved in the conduct of the diagnostic or therapeutic procedure, should administer midazolam. Administering personnel must be trained in the detection and management of airway obstruction, hypoventilation, and apnea, including the maintenance of a patent airway, supportive ventilation, and cardiovascular resuscitation. Resuscitative drugs, and age- and size-appropriate equipment for bag/valve/mask assisted ventilation must be immediately available during administration of midazolam. Continuously monitor vital signs during sedation and during the recovery period.

Syrup:

Midazolam syrup has been associated with respiratory depression and respiratory arrest, especially when used for sedation in noncritical care settings. Midazolam syrup has been associated with reports of respiratory depression, airway obstruction, desaturation, hypoxia, and apnea, most often when used concomitantly with other central nervous system depressants. Midazolam syrup should be used only in hospital or ambulatory care settings, including physicians' and dentists' offices, that can provide for continuous monitoring of respiratory and cardiac function. Immediate availability of resuscitative drugs and age- and size-appropriate equipment for ventilation and intubation, and

personnel trained in their use and skilled in airway management should be assured. For deeply sedated patients, a dedicated individual, other than the practitioner performing the procedure, should monitor the patient throughout the procedure.

Risks from concomitant use with opioids:

Autoinjector, prefilled syringes, vials, and syrup:

Concomitant use of benzodiazepines and opioids may result in profound sedation, respiratory depression, coma, and death. Monitor patients for respiratory depression and sedation.

Premixed IV solutions:

Concomitant use of benzodiazepines and opioids may result in profound sedation, respiratory depression, coma, and death. Continuously monitor patients for respiratory depression and depth of sedation.

Nasal solution:

Concomitant use of benzodiazepines and opioids may result in profound sedation, respiratory depression, coma, and death. Reserve concomitant prescribing of these drugs for patients for whom alternative treatment options are inadequate. Limit dosages and durations to the minimum required. Follow patients for signs and symptoms of respiratory depression and sedation.

Abuse, misuse, and addiction (autoinjector and nasal solution only)

The use of benzodiazepines, including midazolam, exposes users to risks of abuse, misuse, and addiction, which can lead to overdose or death. Abuse and misuse of benzodiazepines commonly involve concomitant use of other medications, alcohol, and/or illicit substances, which is associated with an increased frequency of serious adverse outcomes. Before prescribing midazolam and throughout treatment, assess each patient's risk for abuse, misuse, and addiction.

Dependence and withdrawal reactions (autoinjector and nasal solution only)

The continued use of benzodiazepines may lead to clinically significant physical dependence. The risks of dependence and withdrawal increase with longer treatment duration and higher daily dose. Although midazolam is indicated only for intermittent use, if used more frequently than recommended, abrupt discontinuation or rapid dosage reduction of midazolam may precipitate acute withdrawal reactions, which can be life-threatening. For patients using midazolam more frequently than recommended, to reduce the risk of withdrawal reactions, use a gradual taper to discontinue midazolam.

Individualization of dosage (prefilled syringes and vials only):

Midazolam must never be used without individualization of dosage. The initial intravenous dose for sedation in adult patients may be as little as 1 mg, but should not exceed 2.5 mg in a healthy adult. Lower doses are necessary for older (>60 years of age) or debilitated patients and in patients receiving concomitant opioids or other CNS depressants. The initial dose and all subsequent doses should always be titrated slowly; administer over at least 2 minutes and allow an additional 2 or more minutes to fully evaluate the sedative effect. The use of the 1 mg/mL formulation or dilution of the 1 mg/mL or 5 mg/mL formulation is recommended to facilitate slower injection. Doses of sedative medications in pediatric patients must be calculated on a mg/kg basis, and initial doses and all subsequent doses should always be titrated slowly. The initial pediatric dose of midazolam for sedation/anxiolysis/amnesia is age, procedure, and route dependent.

Neonates (prefilled syringes and vials only):

Midazolam should not be administered by rapid injection in the neonatal population. Severe hypotension and seizures have been reported following rapid IV administration, particularly with concomitant use of fentanyl.

Brand Names: US

- Nayzilam

Brand Names: Canada

- Buccolam

Pharmacologic Category

- Antiseizure Agent, Benzodiazepine;
- Benzodiazepine

Dosing: Adult

Agitation, acute/severe

Agitation, acute/severe (monotherapy or adjunctive therapy) (off-label use):

IV, IM: Initial: 2.5 to 5 mg; repeat doses may be administered every 3 to 5 minutes (IV route) or every 5 to 10 minutes (IM route) until sedation is adequate and appropriate; some patients may require a total dose of ~20 mg; monitor respiratory status; may give alone or in combination with an antipsychotic (Ref).

General anesthesia or monitored anesthesia care

General anesthesia or monitored anesthesia care:

Note: Individualize dosing based on patient factors and concomitant anesthesia.

Premedication/preinduction: Note: Generally, avoid routine administration as a premedication for general anesthesia due to adverse effects (Ref).

IV: Usual dosing range: 0.5 to 2 mg; administered in 0.5 to 1 mg increments based on clinical effect (Ref).

Induction (adjunctive agent): Note: Consider reducing dose or avoiding in hemodynamically unstable patients or if already administered with premedication/preinduction. Generally, avoid routine administration for induction of general anesthesia due to adverse effects (Ref).

IV: Usual dosing range: 0.5 to 2 mg; administered in 0.5 to 1 mg increments based on clinical effect (Ref).

Note: Use as a sole or co-induction agent is described in the manufacturer's labeling and relevant textbooks (ie, 0.1 to 0.25 mg/kg [premedicated patients] and up to 0.35 mg/kg [unpremedicated patients]) (Ref); however, this dosing is uncommonly used and may prolong recovery.

Intoxication

Intoxication: Cocaine, methamphetamine, and other sympathomimetics (alternative agent) (off-label use):

Note : If IV access is not available, consider IM administration; however, effect will be delayed (~10 minutes or greater) (Ref).

IV, IM: 1 to 5 mg every 3 to 10 minutes as needed for agitation, sedation, hyperthermia, hypertension, and tachycardia until desired symptom control is achieved. Large cumulative doses may be required for some patients; monitor for respiratory depression (Ref).

Mechanically ventilated patients in the ICU, sedation

Mechanically ventilated patients in the ICU, sedation (alternative agent):

Note: Used as part of a multimodal strategy. In general, nonbenzodiazepine sedation is preferred due to risk of prolonged sedation and delirium with continuous benzodiazepine use (eg, >48 hours). Intermittent administration is preferred to avoid drug accumulation and prolonged sedation associated with continuous infusions. Titrate to a light level of sedation (eg, Richmond Agitation Sedation Scale 0 to -2) or clinical effect (eg, ventilator dyssynchrony) (Ref). Refer to institutional policies and procedures.

IV:

Intermittent: Initial: 0.5 to 5 mg **or** 0.01 to 0.05 mg/kg over ≥2 minutes; may repeat at 10- to 15-minute intervals as needed until goal level of sedation is achieved (Ref).

Continuous infusion: Loading dose: 0.5 to 5 mg every 1 to 5 minutes (if needed), followed by 1 to 8 mg/hour **or** 0.01 to 0.1 mg/kg/hour continuous infusion; titrate infusion rate to clinical effect. To prevent high basal continuous infusion rates, consider using additional bolus doses instead to achieve goal level of sedation or clinical effect (Ref).

Note: Consider a daily awakening trial or nurse protocolized sedation; if agitated after discontinuation of continuous infusion, then restart at ~50% of the previous dose (Ref).

Palliative and end-of-life sedation

Palliative and end-of-life sedation (off-label use):

Note : Generally used for management of refractory symptoms (eg, anxiety, agitation, palliative sedation) in addition to an opioid and other medications; use in this setting should be done in close consult with an experienced palliative care provider. Ensure that flumazenil is readily available in the case of inadvertent overdose (Ref).

Intermittent dosing:

IV: Initial: 1 to 5 mg; may repeat dose every 5 minutes as needed for symptom management (Ref).

SUBQ: Initial: 2.5 to 10 mg; may repeat dose every 20 minutes as needed for symptom management (Ref).

Continuous infusion: Note: Generally used after failure of intermittent therapy.

IV, SUBQ: Initial: 0.25 to 1 mg/hour; titrate as needed for symptom management; usual dosage range: 0.5 to 5 mg/hour. Some patients may need up to 20 mg/hour; however, before exceeding 10 mg/hour, consider adding another agent (eg, an antipsychotic) or switching to another agent (Ref).

Procedural anxiety

Procedural anxiety (premedication) (off-label use [intranasal and oral]):

IM: 0.07 to 0.08 mg/kg. **Note:** Reserve for situations where other routes are not available or tolerated (Ref).

IV: Initial: 0.5 to 2 mg **or** 0.01 to 0.02 mg/kg over 2 to 5 minutes (maximum single dose: 2 mg); if needed, may administer a repeat dose of 0.25 to 1 mg **or** 0.005 to 0.01 mg/kg approximately 5 to 30 minutes after initial dose (Ref). **Note:** May consider use when immediate relief of anxiety is desired and IV access and monitoring are available (eg, prior to a surgical procedure) (Ref).

Intranasal:

Injectable solution (off-label route):

Note: The marketed intranasal formulation dispenses 5 mg per spray and has not been studied for this indication. Use 5 mg/mL injectable solution to deliver dose (Ref). Due to the low pH of the solution, a burning sensation upon administration is likely to occur; consider using an atomizer for delivery or instilling intranasal lidocaine prior to administration (Ref).

Intranasal: 1 to 4 mg (Ref).

Oral (off-label route) (2 mg/mL oral syrup): 7.5 mg; if needed, may administer a repeat dose of 3.75 mg ~30 to 60 minutes after initial dose, based on incomplete response and/or duration of procedure (Ref).

Procedural sedation, outside the operating room

Procedural sedation, outside the operating room (alternative agent):

IV: Usual dosage range: 0.5 to 2.5 mg over 1 to 2 minutes; repeat every 2 to 5 minutes as needed; titrate to clinical effect; usual total dose: 5 mg (Ref). **Note:** Reduce dose when given in conjunction with opioids or other CNS depressants.

Intranasal (using injectable solution) (off-label route): 0.1 mg/kg (maximum single dose: 10 mg [based on volume]); administer 15 minutes prior to procedure (Ref). **Note:** Use 5 mg/mL injectable solution to deliver dose and administer half the total dose in each nare (Ref). Due to the low pH of the solution, a burning sensation upon administration is likely to occur; consider using an atomizer for delivery or instilling intranasal lidocaine prior to administration (Ref). Some experts may use in combination with intranasal fentanyl (Ref).

Rapid sequence intubation outside the operating room

Rapid sequence intubation outside the operating room (alternative agent) (off-label use):

Note: May cause hypotension; consider alternative agent in patients who are hemodynamically unstable and/or patients with head injury. When used as monotherapy, the onset of action may be up to 5 minutes, which may be too prolonged in some clinical situations (Ref).

IV: 0.2 mg/kg once; use 0.1 mg/kg in hemodynamically unstable patients; usual dosage range: 0.1 to 0.3 mg/kg (Ref).

Seizures

Seizures:

Intermittent (repetitive or cluster): Note: Can be administered by non-health care professionals to patients who are actively seizing or during a seizure cluster. In patients at increased risk of respiratory depression, the first dose should be administered under health care supervision to assess response and tolerability.

Intranasal (Nayzilam nasal spray): 5 mg (1 spray) as a single dose in 1 nostril; may repeat same dose in 10 minutes in alternate nostril based on response and tolerability (do not repeat if the patient is having trouble breathing or excessive sedation). Maximum dose: 10 mg (2 sprays) per single episode.

Maximum treatment frequency: Treatment of 1 episode every 3 days and treatment of 5 episodes in 1 month.

Status epilepticus (convulsive and nonconvulsive):

Note: IM midazolam is the preferred benzodiazepine in patients *without* IV access. Intranasal and buccal administration are effective alternatives in patients *without* IV access (Ref).

IM: 10 mg once or 0.2 mg/kg once (maximum dose: 10 mg) (Ref). **Note:** Only the 10 mg per 0.7 mL autoinjector is FDA approved for status epilepticus; however, any parenteral product approved for IM administration may be used off label for this indication.

Intranasal (may use injectable solution) (off-label route): Initial: 5 to 10 mg. May repeat dose if needed; maximum total dose: 15 mg (Ref). **Note:** May use 5 mg/mL parenteral concentrated solution to deliver dose (Ref). Due to the low pH of the solution, a burning sensation upon administration is likely to occur; consider using an atomizer for delivery or instilling intranasal lidocaine prior to administration (Ref).

Buccal (may use injectable solution) (off-label route): Initial: 10 mg. May repeat dose if needed; maximum total dose: 20 mg (Ref).

Status epilepticus, refractory (off-label use):

Note : Mechanical ventilation and hemodynamic support generally required; titrate dose to cessation of electrographic seizures or burst suppression (Ref).

IV:

Loading dose: 0.2 mg/kg administered slowly, followed by a continuous infusion; repeat loading dose every 3 to 5 minutes (maximum total dose: 2 mg/kg) as needed for electrographic/burst suppression (Ref).

Continuous infusion: After initial loading dose, begin continuous infusion at 0.05 to 2 mg/kg/hour and titrate to cessation of electrographic seizures/burst suppression. For breakthrough status epilepticus, administer bolus of 0.1 to 0.2 mg/kg every 3 to 5 minutes in addition to increasing infusion rate by 0.05 to 0.1 mg/kg/hour every 3 to 4 hours (Ref). Some experts use doses up to 3 mg/kg/hour (Ref).

Note: Generally, a period of at least 24 hours of electrographic suppression is suggested prior to down titrating the continuous infusion; withdraw gradually by decreasing the dose 50% every 3 hours to prevent recurrent status epilepticus or taper off over 4 to 6 hours (Ref).

Substance withdrawal

Substance withdrawal:

Note: Longer-acting benzodiazepines are generally preferred; may be considered when alternative agents are not available (Ref).

Alcohol withdrawal syndrome (alternative agent) (off-label use):

IV: For severe agitation (eg, Clinical Institute Withdrawal Assessment ≥ 16 or Richmond Agitation-Sedation Scale ≥ 3), administer 2 to 5 mg once. If target sedation not achieved, administer 2 to 5 mg every 5 minutes for 3 doses, followed by 4 to 10 mg every 5 minutes for 3 doses (Ref). Maximum total dose: 50 mg.

Opioid withdrawal (adjunctive therapy) (alternative agent) (off-label use): **IV:** 2 mg every 5 to 10 minutes until adequate sedation and hemodynamic stability are achieved (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Note: Use with caution in kidney dysfunction and/or those receiving renal replacement therapies. The half-life of midazolam and active metabolites are prolonged in acutely ill patients with acute kidney injury receiving continuous infusions and may accumulate, contributing to prolonged sedation (Ref).

Altered kidney function:

Buccal, intranasal, parenteral: No initial dosage adjustments necessary for any degree of kidney impairment; however, use with caution and monitor closely for excessive sedation in patients with severe impairment (eg, CrCl < 30 mL/minute); consider longer dosing intervals for intermittent dosing and slower titration of continuous infusions. In some situations (eg, sedation in mechanically ventilated patients) additional boluses may be considered to prevent higher basal continuous infusion rates (Ref).

Hemodialysis, intermittent (thrice weekly): Buccal, intranasal, parenteral: Unlikely to be significantly dialyzable (highly protein bound) (Ref):

No initial dosage adjustments necessary; however, use with caution and monitor closely for excessive sedation; consider longer dosing intervals for intermittent dosing and slower titration of continuous infusions. In some situations (eg, sedation in mechanically ventilated patients), additional boluses may be considered to prevent higher basal continuous infusion rates (Ref).

Peritoneal dialysis: Unlikely to be significantly dialyzable (highly protein bound) (Ref):

Buccal, intranasal, parenteral: No initial dosage adjustments necessary; however, use with caution and monitor closely for excessive sedation; consider longer dosing intervals for intermittent dosing and slower titration of continuous infusions. In some situations (eg, sedation in mechanically ventilated patients) additional boluses may be considered to prevent higher basal continuous infusion rates (Ref).

CRRT: Buccal, intranasal, parenteral: Midazolam and 1-hydroxy-midazolam (active metabolite) are not effectively removed by CRRT, although 1-alpha hydroxymidazolam glucuronide (also hypothesized to be active) is significantly removed (Ref). No initial dosage adjustments necessary; however, use with caution and monitor closely for excessive sedation; consider longer dosing intervals for intermittent dosing and slower titration of continuous infusions. In some situations (eg, sedation in mechanically ventilated patients) additional boluses may be considered to prevent higher basal continuous infusion rates (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration): Buccal, intranasal, parenteral: Midazolam and 1-hydroxymidazolam are not expected to be effectively removed by PIRRT, although 1-alpha hydroxymidazolam glucuronide (also hypothesized to be active) may be significantly removed (Ref). No initial dosage adjustments necessary; however, use with caution and monitor closely for excessive sedation; consider longer dosing intervals for intermittent dosing and slower titration of continuous infusions. In some situations (eg, sedation in mechanically ventilated patients), additional boluses may be considered to prevent higher basal continuous infusion rates (Ref).

Dosing: Liver Impairment: Adult

The liver dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCPS, FCCP, FAST; Arun Jesudian, MD; Sasan Sakiani, MD.

Note: The elimination half-life and clearance of midazolam and its active metabolites are significantly prolonged in patients with cirrhosis (Ref).

Liver impairment prior to treatment initiation:

Initial or dose titration in patients with preexisting liver cirrhosis:

Parenteral: Note: Use of shorter acting agents with no active metabolites (eg, propofol) are preferred over midazolam in patients with cirrhosis (Ref).

Child-Turcotte-Pugh class A to C:

Intermittent dosing: Use lowest effective dose; avoid repeat dosing if possible (Ref).

Continuous infusion: Avoid use; use only if benefits outweigh risks; significant neurocognitive impairment has been observed and may be prolonged post-infusion (Ref).

Buccal, intranasal:

Child-Turcotte-Pugh class A to C: Use should generally be avoided. If use deemed necessary, use the lowest effective dose; avoid repeat dosing if possible (Ref).

Dosing: Obesity: Adult

The recommendations for dosing in patients with obesity are based upon the best available evidence and clinical expertise. Senior Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.

Class 1, 2, and 3 obesity (BMI ≥ 30 kg/m²):

Intermittent:

IV, IM, SUBQ, intranasal (using injectable solution): Use of non-weight-based dosing is preferred; titrate to clinical effect if needed (expert opinion). If weight-based doses are utilized, use **adjusted body weight** for initial weight-based dosing calculations, then titrate to clinical effect if needed (Ref). If rapid sedation or clinical effect is needed, use **actual body weight** for initial weight-based dose calculations, then titrate to clinical effect (Ref). Clinicians should **not** change dosing weight from one weight metric to another during therapy (ie, actual body weight to/from adjusted body weight) (Ref). Refer to adult dosing for indication-specific doses.

Continuous infusion:

IV: Use of non-weight-based dosing is preferred; titrate to clinical effect (Ref). If weight-based doses are utilized, use **adjusted body weight** for initial weight-based dosing calculations, then titrate to clinical effect (Ref). Clinicians should **not** change dosing weight from one weight metric to another during therapy (ie, actual body weight to/from adjusted body weight) (Ref). Refer to adult dosing for indication-specific dosing.

Rationale for recommendations:

There are limited data on the effects of obesity on midazolam dosing or pharmacokinetics. Midazolam pharmacokinetic studies in subjects with obesity have described an increased V_d with no difference in clearance compared to subjects without obesity. The reported half-life is therefore longer in patients with obesity (Ref). Due to the increased V_d , these pharmacokinetic data suggest larger doses may be required. However, larger doses may increase the adverse drug effects particularly in select populations (eg, elderly or nonintubated patients). Furthermore, since obesity does not influence clearance, there may be a potential for accumulation with repeat dosing or continuous infusions (Ref). In general, smaller doses that can be rapidly titrated to effect are preferred (Ref).

Dosing: Older Adult

Note: Avoid use; may be appropriate for seizure disorders and periprocedural anesthesia (Ref).

Intranasal (nasal spray): Refer to adult dosing; use with caution due to potential prolonged drug exposure.

Parenteral: Refer to adult dosing; midazolam needs to be individualized based on the patient's age, underlying diseases, and concurrent medications. Consider reducing dose by 20% to 50% in elderly, chronically ill, or debilitated patients and those receiving opioids or other CNS depressants (Ref). Titration of doses should also be slower (Ref).

Dosing: Pediatric

(For additional information see "Midazolam: Pediatric drug information")

Dosage guidance:

Dosing: Dosage must be individualized and based on patient's age, underlying diseases, concurrent medications, and desired effect; calculate dose based on ideal body weight for patients with obesity; allow 3 to 5 minutes between IV doses to decrease the chance of oversedation.

Dosage form information: The nasal spray formulation delivers a fixed dose of 5 mg and is not appropriate for all pediatric patients; for smaller intranasal doses, parenteral solution for injection can be used; ensure appropriate product selection and administration technique.

Sedation, anxiolysis, and amnesia prior to procedure or before induction of anesthesia

Sedation, anxiolysis, and amnesia prior to procedure or before induction of anesthesia:

IM: Infants, Children, and Adolescents: IM: Usual dose: 0.1 to 0.15 mg/kg; maximum total dose: 10 mg. **Note:** Doses up to 0.5 mg/kg have been used in more anxious patients.

IV:

Infants <6 months: IV: Limited data available in nonintubated infants; infants <6 months are at higher risk for airway obstruction and hypoventilation; titrate dose with small increments to desired clinical effect; monitor carefully.

Infants ≥6 months to Children <6 years: IV: Initial: 0.05 to 0.1 mg/kg; titrate dose carefully to desired clinical effect; total dose of 0.6 mg/kg may be required; usual maximum total dose: 6 mg.

Children ≥6 years: IV: Initial: 0.025 to 0.05 mg/kg; titrate dose carefully to desired clinical effect; total doses of 0.4 mg/kg may be required; usual maximum total dose: 10 mg.

Adolescents: IV: Initial: 1 to 2.5 mg; titrate dose carefully to desired clinical effect; usual maximum total dose: 10 mg. **Note:** Based on experience in adults, smaller initial doses (eg, 0.5 mg) may be considered (Ref).

Intranasal: Limited data available: **Note:** Administer using parenteral solution for injection product via intranasal route; use of mucosal atomizer device (MAD) is recommended.

Infants, Children, and Adolescents: Intranasal: Usual dose: 0.2 mg/kg as a single dose, may repeat in 15 minutes; reported range: 0.2 to 0.8 mg/kg; maximum dose: 10 mg/dose (Ref). **Note:** Some investigators suggest premedication with intranasal lidocaine to decrease irritation and subsequent agitation (Ref).

Oral: **Note:** Individualize dose based on patient status and desired effect.

Infants ≥ 6 months, Children, and Adolescents < 16 years: Oral: Single dose: 0.25 to 0.5 mg/kg once; maximum dose: 20 mg; **Note:** Younger patients (6 months to < 6 years) and those less cooperative may require higher doses (up to 1 mg/kg) compared to patients 6 years to < 16 years, who may only require 0.25 mg/kg; use lower initial doses (0.25 mg/kg) in all patients with cardiac or respiratory compromise, concomitant CNS depressant, or high-risk surgical patients (Ref).

Adolescents ≥ 16 years: Oral: Single dose: 0.25 to 0.5 mg/kg once up to a maximum dose of 20 mg/dose; usual adult dose: 7.5 mg/dose. **Note:** Use lower initial doses (0.25 mg/kg) in all patients with cardiac or respiratory compromise, concomitant CNS depressant, or high-risk surgical patients (Ref).

Rectal: Limited data available: Infants > 6 months and Children: Rectal: Usual: 0.25 to 0.5 mg/kg once (Ref); doses up to 1 mg/kg have been used in infants and young children (7 months to 5 years of age) but may be associated with a higher incidence of postprocedural agitation (Ref).

Sedation, mechanically ventilated patient

Sedation, mechanically ventilated patient: Infants, Children, and Adolescents: IV: Loading dose: 0.05 to 0.2 mg/kg given slow IV over ≥ 2 to 3 minutes, then follow with initial continuous IV infusion: 0.05 to 0.12 mg/kg/hour (0.8 to 2 mcg/kg/minute); titrate to the desired effect; maximum initial dose: 10 mg/hour (Ref).

Sedation tapering (after prolonged therapy): After prolonged therapy, consider a slow taper of therapy or conversion to a long-acting benzodiazepine to prevent signs and symptoms of withdrawal. Reported conversion strategies are based on potency, half-life, and oral bioavailability of diazepam and lorazepam but are variable and based on limited clinical evidence (Ref).

If duration of therapy < 3 to 5 days: Decrease dose by 10% to 15% every 6 to 8 hours (Tobias 2000); monitor closely for signs and symptoms of withdrawal with each reduction in dose.

If duration of therapy > 5 days, the following regimens have been reported:

Conversion to oral diazepam: After first dose of oral diazepam, decrease midazolam infusion by 50%; after the second dose of oral diazepam, discontinue the midazolam infusion (Ref); monitor closely for signs and symptoms of withdrawal with each reduction in dose.

Conversion to oral lorazepam: After second dose of oral lorazepam, decrease midazolam infusion by 50%; after third dose of oral lorazepam, decrease the midazolam infusion by an additional 50%; after fourth dose of oral lorazepam, discontinue the midazolam infusion (Ref); monitor closely for signs and symptoms of withdrawal with each reduction in dose.

Seizures, acute treatment

Seizures, acute treatment:

Buccal: Limited data available (Ref): **Note:** Reserve for patients without IV access; if seizure does not cease within 5 minutes, some studies describe repeating dose (Ref):

Weight-directed dosing: Infants ≥ 3 months, Children, and Adolescents: Buccal: 0.2 to 0.5 mg/kg once; maximum dose: 10 mg/dose.

Age-directed dosing:

Infants ≥ 3 months: Buccal: 2.5 mg.

Children 1 to 4 years: Buccal: 5 mg.

Children 5 to 9 years: Buccal: 7.5 mg.

Children and Adolescents ≥ 10 years: Buccal: 10 mg.

IM: Limited data available:

Weight-directed dosing: Infants, Children, and Adolescents: IM: 0.2 mg/kg/dose (Ref).

Fixed dosing (Ref):

13 to 40 kg: IM: 5 mg.

>40 kg: IM: 10 mg.

Intranasal:

Nasal spray (Nayzilam): Children ≥ 12 years and Adolescents: Intranasal: 5 mg administered as 1 spray into 1 nostril; may repeat dose in 10 minutes in alternate nostril based on response and tolerability; do not repeat dose if patient has difficulty breathing or excessive sedation; maximum dose: 10 mg/dose per episode (2 sprays); do not exceed maximum treatment frequency of 1 episode every 3 days and 5 episodes per month.

Parenteral solution for injection product: Limited data available: Use of mucosal atomizer device (MAD) is recommended:

Infants, Children, and Adolescents: Intranasal (using parenteral solution): 0.2 mg/kg/dose, divide dose between nares; maximum dose: 10 mg/dose; may repeat once to a total maximum of 0.4 mg/kg (Ref). **Note:** A higher dose of 0.3 mg/kg has been described in patients ≥ 6 months (Ref).

Status epilepticus

Status epilepticus:

Standard treatment: Infants, Children, and Adolescents: Limited data available:

IM:

Weight-based dosing: IM: 0.2 mg/kg once; maximum dose: 10 mg/dose (Ref).

Fixed dosing (Ref):

13 to 40 kg: IM: 5 mg once.

>40 kg: IM: 10 mg once.

Intranasal: 0.2 mg/kg once; maximum dose: 10 mg/dose (Ref).

Buccal: 0.5 mg/kg once; maximum dose: 10 mg/dose (Ref).

Refractory to standard treatment (Ref): **Note:** Mechanical ventilation and cardiovascular monitoring required; titrate dose to cessation of electrographic seizures or burst suppression. Infants, Children, and Adolescents: Limited data available:

Loading dose: IV: 0.2 mg/kg followed by a continuous infusion.

Continuous IV infusion: 0.05 to 2 mg/kg/hour (0.83 to 33.3 mcg/kg/minute) titrated to cessation of electrographic seizures or burst suppression. If patient experiences breakthrough status epilepticus while on the continuous infusion, administer a bolus of 0.1 to 0.2 mg/kg and increase infusion rate by 0.05 to 0.1 mg/kg/hour (0.83 to 1.66 mcg/kg/minute) every 3 to 4 hours.

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling; use with caution; half-life of midazolam and metabolites may be prolonged. Adult patients with renal failure receiving

a continuous infusion cannot adequately eliminate the active hydroxylated metabolites (eg, 1-hydroxymidazolam) contributing to prolonged sedation sometimes for days after discontinuation (Ref).

Hemodialysis: Supplemental dose is not necessary.

Peritoneal dialysis: Significant drug removal is unlikely based on physiochemical characteristics

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling; use with caution. Based on experience in adult patients, the following have been suggested and may be considered in pediatric patients:

Single dose (eg, induction): No dosage adjustment recommended; patients with hepatic impairment may be more sensitive compared to patients without hepatic impairment; anticipate longer duration of action (Ref).

Multiple dosing or continuous infusion: Expect longer duration of action and accumulation; based on patient response, dosage reduction likely to be necessary (Ref).

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. As reported in children, adolescents, and adults unless otherwise noted. Intranasal-only adverse reactions include adolescents and adults. Oral-only adverse reactions include children and adolescents.

>10%:

Gastrointestinal: Vomiting ($\leq 11\%$)

Respiratory: Apnea (children: 3%; adults: 15%), bradypnea (adults: $\leq 23\%$), decreased tidal volume (adults: $\leq 23\%$), nasal discomfort (intranasal: 5% to 16%)

1% to 10%:

Cardiovascular: Atrioventricular nodal arrhythmia ($\leq 1\%$), bigeminy ($\leq 2\%$), bradycardia ($\leq 2\%$), hypotension ($\leq 3\%$), premature ventricular contractions ($\leq 1\%$), syncope ($\leq 1\%$), tachycardia ($\leq 1\%$)

Dermatologic: Pruritus ($\leq 1\%$), skin rash ($\leq 2\%$), urticaria ($\leq 1\%$)

Gastrointestinal: Acidic taste ($\leq 1\%$), dysgeusia (intranasal: 4%), hiccups (1% to 4%), nausea (2% to 5%), retching ($\leq 1\%$), sialorrhea ($\leq 1\%$)

Hematologic & oncologic: Hematoma ($\leq 1\%$), oxygen desaturation ($\leq 5\%$)

Hypersensitivity: Anaphylaxis ($\leq 1\%$), hypersensitivity reaction ($\leq 1\%$)

Local: Burning sensation at injection site ($\leq 1\%$), erythema at injection site (adults: $\leq 3\%$), induration at injection site (adults: $\leq 2\%$), injection-site reaction (coldness: $\leq 1\%$), pain at injection site (adults: 4% to 5%), swelling at injection site ($\leq 1\%$), tenderness at injection site (adults: 6%), urticaria at injection site ($\leq 1\%$), warm sensation at injection site ($\leq 1\%$)

Nervous system: Abnormal dreams (emergence: 1%), agitation ($\leq 4\%$; including emergence agitation), anxiety ($\leq 1\%$), asthenia ($\leq 1\%$), ataxia ($\leq 1\%$), athetosis ($\leq 1\%$), behavioral changes (argumentativeness: $\leq 1\%$), chills ($\leq 1\%$), confusion ($\leq 1\%$), delirium (emergence: $\leq 1\%$), dizziness ($\leq 1\%$), drowsiness (injection [adults]: 1%; intranasal: 9% to 10%), dysarthria (intranasal: 2%), dysphoria ($\leq 1\%$), euphoria ($\leq 1\%$), hallucination ($\leq 1\%$), headache (adolescents, adults: 1% to 7%), impaired consciousness (adults: 3%), insomnia ($\leq 1\%$), intoxicated feeling ($\leq 1\%$), lethargy ($\leq 1\%$), loss of balance ($\leq 1\%$), mental status changes (adults: 3%), nervousness ($\leq 1\%$), nightmares ($\leq 1\%$), paresthesia ($\leq 1\%$), prolonged emergence from anesthesia ($\leq 1\%$), psychiatric signs and symptoms (oral: $\leq 3\%$), restlessness ($\leq 1\%$), retrograde amnesia ($\leq 1\%$), sedated state (prolonged: oral: 2%; severe sedation [adults]: 2%), seizure-like activity ($\leq 1\%$), sleep disturbance ($\leq 1\%$), slurred speech ($\leq 1\%$), voice disorder ($\leq 1\%$), yawning ($\leq 1\%$)

Neuromuscular & skeletal: Laryngospasm ($\leq 4\%$)

Ophthalmic: Accommodation disturbance ($\leq 1\%$), blurred vision ($\leq 1\%$), diplopia ($\leq 1\%$), increased lacrimation (intranasal: 1% to 2%), miosis ($\leq 1\%$), nystagmus disorder ($\leq 1\%$), visual disturbance ($\leq 1\%$)

Otic: Blockage of external ear ($\leq 1\%$)

Renal: Acute kidney injury (adults: 2%)

Respiratory: Airway obstruction ($\leq 5\%$), bronchospasm ($\leq 1\%$), cough (adults: 1%), dyspnea ($\leq 1\%$), hyperventilation ($\leq 1\%$), hypoxia (oral: 3% to 4%), respiratory congestion (upper; oral: 2%), respiratory depression ($\leq 2\%$), rhinorrhea (intranasal, oral: $\leq 5\%$), rhonchi (children: $\leq 2\%$), shallow respiration ($\leq 1\%$), tachypnea ($\leq 1\%$), throat irritation (intranasal: 2% to 7%), wheezing ($\leq 1\%$)

Miscellaneous: Fever (adults: 4%), paradoxical reaction (children: 2%)

$<1\%$:

Local: Injection-site phlebitis

Neuromuscular & skeletal: Muscle rigidity

Respiratory: Sneezing

Postmarketing: Nervous system: Cognitive dysfunction, drug dependence (physical and psychological dependence with prolonged use), hyperactive behavior, involuntary motor activity, suicidal ideation, suicidal tendencies, tonic-clonic movements, tremor

Contraindications

Injection, oral: Hypersensitivity to midazolam or any component of the formulation; intrathecal or epidural injection of parenteral forms containing preservatives (ie, benzyl alcohol); use in premature infants for parenteral forms containing benzyl alcohol; acute narrow-angle glaucoma (excluding autoinjector).

Intranasal: Hypersensitivity to midazolam or any component of the formulation; acute narrow-angle glaucoma.

Canadian labeling: Additional contraindications (not in the US labeling):

Buccal oral solution: Myasthenia gravis; acute pulmonary insufficiency; sleep apnea syndrome; severe chronic obstructive pulmonary disease (COPD); severe hepatic impairment.

Injection: Acute pulmonary insufficiency; severe COPD; IV sedation in elderly or debilitated patients outside ICU setting and in patients not sufficiently alert to respond appropriately to verbal requests.

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Warnings/Precautions

Concerns related to adverse effects:

- Anterograde amnesia: Benzodiazepines have been associated with anterograde amnesia (Nelson 1999).
- Cardiorespiratory effects: Serious cardiorespiratory adverse events have occurred with midazolam administration, including cardiac arrest, permanent neurologic injury, and death. Risk of adverse events is increased in patients with abnormal airway anatomy, cyanotic congenital heart disease, sepsis, or severe pulmonary disease. In patients with a risk of respiratory depression, consider administering the first dose of intranasal midazolam under health care supervision; this may be performed in the absence of a seizure episode.
- CNS depression: May cause CNS depression, which may impair physical or mental abilities; patients must be cautioned about performing tasks that require mental alertness (eg, operating machinery, driving). A minimum of one day should elapse after midazolam administration before attempting

these tasks. Elapsed time to resume these tasks must be individualized, as pharmacologic effects are dependent on dose, route, duration of procedure, and presence of other medications.

- Hypotension: May cause hypotension, particularly in pediatric patients or patients with hemodynamic instability. Hypotension may occur more frequently in patients who have received opioid analgesics.
- Paradoxical reactions: Paradoxical reactions, including hyperactive or aggressive behavior, have been reported with benzodiazepines; risk may be increased in adolescent/pediatric patients, older adults, or patients with a history of alcohol use disorder or psychiatric/personality disorders (Mancuso 2004). Midazolam may cause involuntary movements (eg, tonic/clonic movements, tremor) and combativeness when used for sedation; may cause agitation when used for sedation or status epilepticus. Reactions may be due to improper dosing or administration; cerebral hypoxia should also be considered as a cause.
- Suicidal ideation: Intranasal: Pooled analysis of trials involving various antiseizure medications (regardless of indication) showed an increased risk of suicidal thoughts/behavior (incidence rate: 0.43% treated patients compared to 0.24% of patients receiving placebo); risk observed as early as one week after initiation and continued through duration of trials (most trials ≤ 24 weeks). Monitor all patients for notable changes in behavior that might indicate suicidal thoughts or depression; notify health care provider immediately if symptoms occur.

Disease-related concerns:

- Acute illness: Use IV midazolam with caution in patients with uncompensated acute illnesses, such as severe fluid or electrolyte disturbances.
- Cardiovascular disease: Use with caution in patients with heart failure. Adverse hemodynamic events have been reported in pediatric patients with cardiovascular instability; avoid rapid IV administration in these patients.
- Glaucoma: Use with caution in patients with glaucoma; may increase intraocular pressure. Some formulations are contraindicated in acute narrow-angle glaucoma. May consider use in patients with open-angle glaucoma only if receiving appropriate therapy; consider evaluating ophthalmologic status after midazolam use.
- Renal impairment: Use with caution in patients with renal impairment; half-life of midazolam and metabolites may be prolonged.
- Respiratory disease: Reduce dose or avoid use in patients with respiratory disease, including chronic obstructive pulmonary disease or sleep apnea. Benzodiazepines may cause significant respiratory depression.

Special populations:

- Debilitated patients: Use with caution in debilitated patients; decreased dosages recommended. These patients take longer to recover completely after midazolam administration for the induction of anesthesia.
- Fall risk: Use with extreme caution in patients who are at risk of falls; benzodiazepines have been associated with falls and traumatic injury (Nelson 1999).
- Neonates: Injection: Neonates are vulnerable to profound and/or prolonged respiratory effects of midazolam.
- Obesity: Use benzodiazepines with caution in patients with obesity; may have prolonged action when discontinued.
- Older adults: Use with caution in older adults; decreased dosages and slower titration is recommended due to an increased volume of distribution seen with lipophilic drugs in older adults, resulting in slower distribution and lower clearance. The risk of hypoventilation, airway obstruction, and apnea is higher in older patients. Older adults can also take longer to recover completely after midazolam administration for the induction of anesthesia (Strøm 2016). Use of oral midazolam is not recommended in older adults. Older adults may be at an increased risk of death with use; risk has been

found highest within the first 4 months of use in older patients with dementia (Jennum 2015; Saarelainen 2018).

- **Pediatric:** Pediatric patients with cardiac or respiratory compromise may be sensitive to the respiratory depressant effect of midazolam. Pediatric patients undergoing procedures involving the upper airway (eg, upper endoscopy, dental care) are vulnerable to episodes of desaturation and hypoventilation.
- **Pediatric neurotoxicity:** In pediatric and neonatal patients <3 years of age and patients in third trimester of pregnancy (ie, times of rapid brain growth and synaptogenesis), the repeated or lengthy exposure to sedatives or anesthetics during surgery/procedures may have detrimental effects on child or fetal brain development and may contribute to various cognitive and behavioral problems. Epidemiological studies in humans have reported various cognitive and behavioral problems, including neurodevelopmental delay (and related diagnoses), learning disabilities, and attention-deficit hyperactivity disorder. Human clinical data suggest that single, relatively short exposures are not likely to have similar negative effects. No specific anesthetic/sedative has been found to be safer. For elective procedures, risk versus benefits should be evaluated and discussed with parents/caregivers/patients; critical surgeries should not be delayed (US FDA Safety Communication 2017 Update).

Dosage form specific issues:

- **Benzyl alcohol and derivatives:** Some dosage forms may contain benzyl alcohol; large amounts of benzyl alcohol (≥ 99 mg/kg/day) have been associated with a potentially fatal toxicity ("gasping syndrome") in neonates; the "gasping syndrome" consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension, and cardiovascular collapse (AAP ["Inactive" 1997]; CDC 1982); some data suggest that benzoate displaces bilirubin from protein binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol with caution in neonates. See manufacturer's labeling.

Other warnings/precautions:

- **Abuse, misuse, and substance use disorder:** Counsel patients at increased risk on proper use and monitoring for signs and symptoms of abuse, misuse, and substance use disorder. Institute early treatment or refer patients in whom substance use disorder is suspected. Limit dosages and durations to the minimum required; counsel on proper disposal of unused drug.
- **Appropriate use:** Does not have analgesic, antidepressant, or antipsychotic properties. Does not protect against increases in intracranial pressure, heart rate, and/or blood pressure during intubation. Do not use in shock, coma, or acute alcohol intoxication with depression of vital signs. Avoid intra-arterial administration or extravasation of parenteral formulations. Use during upper airway procedures (ie, endoscopy, dental care) may increase risk of hypoventilation. Prolonged responses have been noted following extended administration by continuous infusion (possibly due to metabolite accumulation) or in the presence of drugs which inhibit midazolam metabolism. Oral midazolam is intended for use in monitored settings only and not for chronic or home use.
- **Dependence and withdrawal reactions:** Some patients may develop a protracted withdrawal syndrome lasting >12 months; may be difficult to differentiate withdrawal symptoms from reemergence or continuation of symptoms for which benzodiazepines were prescribed. Flumazenil may cause withdrawal in patients receiving long-term benzodiazepine therapy.
- **Tolerance:** Midazolam is a short half-life benzodiazepine and may be of benefit in patients where a rapidly and short-acting agent is desired (eg, for acute agitation). Duration of action after a single dose is determined by redistribution rather than metabolism. Tolerance develops to the sedative and antiseizure effects. It does not develop to the anxiolytic effects (Vinkers 2012).

Warnings: Additional Pediatric Considerations

In pediatric and neonatal patients <3 years of age and patients in third trimester of pregnancy (ie, times of rapid brain growth and synaptogenesis), the repeated or lengthy exposure to sedatives or anesthetics during surgery/procedures may have detrimental effects on the child's or fetus' brain development and may contribute to various cognitive and behavioral problems; the FDA is requiring

warnings be included in the manufacturer's labeling for all general anesthetic/sedative drugs. Multiple animal species studies have shown adverse effects on brain maturation; in juvenile animals, drugs that potentiate GABA activity and/or block NMDA receptors for >3 hours demonstrated widespread neuronal and oligodendrocyte cell loss along with alteration in synaptic morphology and neurogenesis. Epidemiological studies in humans have reported various cognitive and behavioral problems including neurodevelopmental delay (and related diagnoses), learning disabilities, and attention-deficit/hyperactivity disorder. Human clinical data suggest that single, relatively short exposures are not likely to have similar negative effects. Further studies are needed to fully characterize findings and ensure that these findings are not related to underlying conditions or the procedure itself. No specific anesthetic/sedative has been found to be safer. For elective procedures, risk vs benefits should be evaluated and discussed with parents/caregivers/patients; critical surgeries should not be delayed (FDA 2017; McCann 2019; Sun 2016).

In neonates, particularly premature neonates, several cases of myoclonus (rhythmic myoclonic jerking) have been reported (~8% incidence).

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Injection:

Generic: 2 mg/2 mL (2 mL); 5 mg/5 mL (5 mL); 10 mg/10 mL (10 mL); 5 mg/mL (1 mL); 10 mg/2 mL (2 mL); 25 mg/5 mL (5 mL); 50 mg/10 mL (10 mL)

Solution, Injection [preservative free]:

Generic: 2 mg/2 mL (2 mL); 5 mg/5 mL (5 mL); 5 mg/mL (1 mL); 10 mg/2 mL (2 mL)

Solution, Intravenous [preservative free]:

Generic: 100 mg/100 mL in NaCl 0.8% (100 mL); 100 mg/100 mL in NaCl 0.9% (100 mL); 50 mg/50 mL in NaCl 0.8% (50 mL); 50 mg/50 mL in NaCl 0.9% (50 mL)

Solution, Nasal:

Nayzilam: 5 mg/0.1 mL (1 ea) [contains propylene glycol]

Solution Auto-injector, Intramuscular [preservative free]:

Generic: 10 mg/0.7 mL (0.7 mL)

Syrup, Oral:

Generic: 2 mg/mL (2.5 mL [DSC], 5 mL [DSC], 118 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Midazolam HCl (PF) Injection)

2 mg/2 mL (per mL): \$0.71 - \$1.88

5 mg/5 mL (per mL): \$0.21 - \$0.88

5 mg/mL (per mL): \$1.13 - \$2.98

10 mg/2 mL (per mL): \$0.64 - \$2.29

Solution (Midazolam HCl Injection)

2 mg/2 mL (per mL): \$0.19 - \$0.78

5 mg/5 mL (per mL): \$0.13 - \$0.78

5 mg/mL (per mL): \$1.44 - \$3.91

10 mg/10 mL (per mL): \$0.26 - \$0.68

10 mg/2 mL (per mL): \$0.75 - \$3.86

25 mg/5 mL (per mL): \$1.20 - \$1.70

50 mg/10 mL (per mL): \$0.23 - \$4.15

Solution (Midazolam-Sodium Chloride (PF) Intravenous)

50 mg/50 mL 0.9% (per mL): \$0.53

50MG/50ML 0.8% (per mL): \$0.19

100 mg/100 mL 0.9% (per mL): \$0.36

100MG/100ML 0.8% (per mL): \$0.10

Solution (Midazolam-Sodium Chloride Intravenous)

50 mg/50 mL 0.9% (per mL): \$0.45 - \$0.53

100 mg/100 mL 0.9% (per mL): \$0.31 - \$0.36

Solution (Nayzilam Nasal)

5MG/0.1ML (per each): \$410.56

Solution Auto-injector (Midazolam Intramuscular)

10 mg/0.7 mL (per 0.7 mL): \$240.00 - \$240.00

Syrup (Midazolam HCl Oral)

2 mg/mL (per mL): \$0.69 - \$2.16

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Buccal:

Buccolam: 2.5 mg/0.5 mL (0.5 mL); 5 mg/mL (1 mL); 7.5 mg/1.5 mL (1.5 mL); 10 mg/2 mL (2 mL)

Solution, Injection:

Generic: 1 mg/mL (2 mL, 5 mL, 10 mL, 50 mL, 100 mL); 5 mg/mL (1 mL, 2 mL, 10 mL, 50 mL); 50 mg/10 mL (10 mL)

Controlled Substance

C-IV

Administration: Adult

Buccal (off-label route): A buccal formulation is not currently available in the US. Some trials used an injectable solution administered buccally (Ref). International studies used a 10 mg/mL commercially available buccal formulation (Ref). Administer to the buccal mucosa between the gums and

the cheek using an oral syringe; gently massage cheek; dose may be divided to both sides of the mouth (Ref).

Intranasal:

Nasal spray: Do not test or prime before use. Administer 1 spray into 1 nostril; if a second dose is needed, administer in alternate nostril. A second dose should not be administered if the patient is having trouble breathing or excessive sedation.

Solution, injection (off-label route): **Note:** Due to the low pH of the solution, burning upon administration is likely to occur (Ref). To reduce irritation, use an atomizer **or** spray 1 mL of lidocaine 2% or 4% delivered 5 minutes prior to administering midazolam (Ref).

Using the 5 mg/mL injectable solution (concentration minimizes volume administered, which reduces irritation and swallowing the dose), draw up desired dose with a 1 to 3 mL syringe; may attach a nasal mucosal atomization device prior to delivering dose. The maximum recommended dose volume (of the 5 mg/mL concentration) per nare is 1 mL. Deliver half of the total dose into the first nare using the atomizer device or by dripping slowly into nostril, then deliver the other half of the dose into the second nare (Ref).

Oral: Do not mix with any liquid (such as grapefruit juice) prior to administration. Administer on empty stomach (feeding is usually contraindicated prior to sedation for procedures).

Parenteral: Do **not** administer intraarterially.

IM: Administer undiluted deep IM into large muscle. The manufacturer's labeling for the 10 mg per 0.7 mL autoinjector specifically recommends injection in the mid-outer thigh (vastus lateralis muscle). May inject autoinjector through clothing. Hold autoinjector firmly in place for at least 10 seconds. Refer to manufacturer's labeling and/or instructions for use for additional information.

IV: For procedural sedation/anxiolysis/amnesia, administer by slow IV injection over at least 2 minutes using a concentration of 1 mg/mL or a dilution of the 1 or 5 mg/mL concentrations. For induction of anesthesia, administer IV bolus over 5 to 15 seconds (Ref). For refractory status epilepticus, the loading dose is recommended to be infused slowly (Ref). For other clinical situations (eg, sedation in the mechanically ventilated patient), a continuous infusion may also be administered.

Rectal: Clinical trials utilized parenteral midazolam for rectal administration; administer a 1 to 5 mg/mL solution through a small, lubricated catheter or tube inserted rectally; hold buttocks closed for ~5 minutes after administration (Ref).

Preparation for Administration: Adult

Note: Premixed solutions are available.

For continuous IV infusion, may dilute with NS or D5W to a final concentration of 0.5 mg/mL or 1 mg/mL (Ref).

Usual Infusion Concentrations: Adult

Note: Premixed solutions available.

IV infusion: 100 mg in 100 mL (concentration: 1 mg/mL) of D₅W or NS.

Administration: Pediatric

Buccal: A buccal formulation is not currently available in the United States. Some trials used an injectable solution administered buccally (Ref). International studies used a 10 mg/mL commercially available buccal formulation (Ref). Administer to the buccal mucosa between the gums and the cheek using an oral syringe; gently massage cheek; dose may be divided to both sides of the mouth (Ref).

Intranasal: The nasal spray formulation (Nayzilam) delivers a fixed dose of 5 mg and is not appropriate for all pediatric patients; for smaller intranasal doses, parenteral solution for injection product can be used; ensure appropriate product selection and administration technique.

Nasal spray (Nayzilam): Product delivers a fixed dose and should only be used in Children ≥12 years and Adolescents: Do not test or prime before use; do not open blister packaging until ready to use.

Administer 1 spray into 1 nostril; if a second dose is needed, administer in alternate nostril. A second dose should not be administered if the patient is having trouble breathing or experiencing excessive sedation.

Parenteral solution for injection formulation: Administer using a mucosal atomizer (MAD nasal drug delivery device) or via needleless syringe. Administer half of the dose into each nostril. Maximum volume: 1 mL per nostril (Ref).

Oral: Administer on empty stomach (feeding is usually contraindicated prior to sedation for procedures).

Parenteral: Vial for IV or IM administration only; premixed bag for IV administration only. Do not administer vial or premixed bag intra-arterially. Avoid extravasation.

IV (bolus, loading doses, intermittent therapy): Administer by slow IV injection over ≥ 2 to 5 minutes (Ref). In adults: For induction of anesthesia, may administer IV push over 20 to 30 seconds per the manufacturer. For refractory status epilepticus, the loading dose is recommended to be infused at a rate of 2 mg/minute (Ref).

Neonates: Rapid administration (< 2 minutes) has been reported to cause severe hypotension especially if administered concurrently with fentanyl. For loading doses, administration over ≥ 1 hour has successfully prevented hypotension in neonates (Ref); however, administration over 2 to 5 minutes has also been described in mechanically ventilated patients (Ref); manufacturer recommends against loading doses in neonates and suggests using a faster continuous infusion rate for the first several hours.

Continuous IV infusion: Administer via an infusion pump.

IM: Administer undiluted deep IM into large muscle, generally into anterior-lateral aspect of thigh (vastus lateralis) in pediatric patients (Ref).

Rectal: Clinical trials utilized parenteral midazolam for rectal administration; administer a 1 to 5 mg/mL solution through a small, lubricated catheter or tube inserted rectally; hold buttocks closed for ~ 5 minutes after administration (Ref).

Preparation for Administration: Pediatric

Parenteral:

Intermittent IV: For procedural sedation/anxiolysis/amnesia, the 1 mg/mL concentration is recommended to facilitate slower injection per the manufacturer; may also dilute the 1 mg/mL and 5 mg/mL formulations with NS or D5W. For neonates, use the preservative-free injection; alternatively, if the preservative-free product is not available, may use the 5 mg/mL injection and dilute to 0.5 mg/mL with SWFI (preservative free) to decrease the amount of benzyl alcohol delivered to the neonate (both concentrations of midazolam injection contain 1% benzyl alcohol). Vermont Oxford Network recommends a standard concentration of 0.2 mg/mL or 1 mg/mL for intermittent doses for neonates (Ref).

Continuous IV infusion: Dilute with NS or D5W to a final concentration of 0.5 mg/mL or 1 mg/mL (Ref); a commercially prepared premixed bag is available in concentration of 1 mg/mL. For neonatal patients, Vermont Oxford Network recommends a standard concentration of 0.2 mg/mL or 1 mg/mL for neonates (Ref). For pediatric patients, ASHP recommends standard concentrations of 0.3 mg/mL, 1 mg/mL, or 5 mg/mL (Ref).

Usual Infusion Concentrations: Pediatric

Note: Premixed solutions available (1 mg/mL).

IV infusion: 0.3 mg/mL, 1 mg/mL, **or** 5 mg/mL.

Storage/Stability

Buccal oral solution [Canadian product]: Store at 15°C to 30°C (59°F to 86°F); do not refrigerate or freeze. Keep syringe in the protective plastic tube.

Oral: Store at 25°C (77°F); excursions permitted to 15°C to 30°C (59°F to 86°F).

Injection:

Autoinjector: Store at 20°C to 25°C (68°F to 77°F); excursions permitted between 15°C and 30°C (59°F and 86°F). Do not freeze.

Prefilled syringe: Store at 20°C to 25°C (68°F to 77°F). A final concentration of midazolam 0.5 mg/mL is stable for up to 24 hours when diluted with D5W or NS or 4 hours when diluted with LR.

Premixed bag: Store at 20°C to 25°C (68°F to 77°F); do not freeze. Discard any portion not used within 48 hours of initial penetration.

Vials: Store intact vials at 20°C to 25°C (68°F to 77°F); excursions permitted between 15°C and 30°C (59°F and 86°F). The product labeling for some products recommends to protect from light; refer to the manufacturer's labeling for specific recommendations. A final concentration of midazolam 0.5 mg/mL is stable for up to 24 hours when diluted with D5W or NS or 4 hours when diluted with LR. A final concentration of 1 mg/mL in NS has been documented to be stable for up to 10 days (McMullin 1995) and up to 27 days in D5W (Karlage 2011).

Intranasal (nasal spray): Store at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C to 30°C (59°F to 86°F). Only open blister pack immediately prior to administration.

Use: Labeled Indications

General anesthesia: Induction of general anesthesia before administration of other anesthetic agents; maintenance of anesthesia as a component of balanced anesthesia.

Mechanically ventilated patients in the ICU, sedation: Sedation of intubated and mechanically ventilated patients as a component of anesthesia or during treatment in a critical care setting by continuous IV infusion.

Procedural sedation, outside the operating room: Sedation, anxiolysis, and amnesia prior to or during diagnostic, therapeutic, or endoscopic procedures, or prior to surgery.

Seizures, intermittent:

Intranasal: Acute treatment of intermittent, stereotypic episodes of frequent seizure activity (ie, seizure clusters, acute repetitive seizures) that are distinct from a patient's usual seizure pattern in patients with epilepsy ≥ 12 years of age.

Buccal oral solution [Canadian product]: Treatment of prolonged (>5 minutes), acute, convulsive seizures distinct from a patient's usual seizure pattern in patients 3 months to 17 years of age with epilepsy.

Status epilepticus: Treatment of status epilepticus in adults. **Note:** Only the 10 mg per 0.7 mL autoinjector is FDA approved for the treatment of status epilepticus; however, any parenteral product approved for IM administration may be used off label for this indication.

Use: Off-Label: Adult

Agitation, acute/severe; Intoxication: Cocaine, methamphetamine, and other sympathomimetics; Palliative and end-of-life sedation; Procedural anxiety (premedication); Rapid sequence intubation, outside the operating room; Status epilepticus (convulsive and nonconvulsive); Status epilepticus, refractory; Substance withdrawal

Medication Safety Issues

Sound-alike/look-alike issues:

Versed may be confused with VePesid, Vistaril

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (moderate sedation agent, IV; moderate and minimal sedation agent, oral, for children; pediatric liquid medications requiring measurement) which have a heightened risk of causing significant

patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024; ISMP List of High-Alert Medications in Community/Ambulatory Care Settings ©ISMP 2021).

Older Adult: High-Risk Medication:

Beers Criteria: Benzodiazepines (eg, midazolam) are identified in the Beers Criteria as potentially inappropriate medications to be avoided in patients 65 years and older because of risk of abuse, misuse, physical dependence, and addiction. In addition, older adults have increased risk of impaired cognition, delirium, falls, fractures, and motor vehicle accidents with benzodiazepine use; however, benzodiazepines may be appropriate in older adults when used for seizure disorders, rapid eye movement sleep behavior disorder, benzodiazepine or ethanol withdrawal, severe generalized anxiety disorder, or procedural anesthesia (Beers Criteria [AGS 2023]).

Midazolam is identified in the Screening Tool of Older Person's Prescriptions (STOPP) criteria as a potentially inappropriate medication in older adults (≥65 years of age) with a history of recurrent falls due to an increased risk of falls. Use of benzodiazepines is not recommended for ≥4 weeks. In addition, some disease states of concern include dementia and respiratory failure (acute or chronic) (O'Mahony 2023).

Pediatric patients: High-risk medication:

KIDs List: Midazolam, when used in neonates weighing <1,500 g, is identified on the Key Potentially Inappropriate Drugs in Pediatrics (KIDs) list and should be used with caution due to risk of severe intraventricular hemorrhage, periventricular leukomalacia, or death (weak recommendation; low quality of evidence) (PPA [McPherson 2025]).

Metabolism/Transport Effects

Substrate of CYP2B6 (Minor), CYP3A4 (Major); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Acrivastine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Alcohol (Ethyl): CNS Depressants may increase CNS depressant effects of Alcohol (Ethyl). *Risk C: Monitor*

Aldesleukin: May increase serum concentrations of CYP Substrates (Narrow Therapeutic Index/Sensitive with Inhibitors). *Risk C: Monitor*

Alizapride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Allopurinol: CNS Depressants may increase sedative effects of Allopurinol. *Risk C: Monitor*

Amisulpride (Oral): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Amitriptyline: CNS Depressants may increase CNS depressant effects of Amitriptyline. *Risk C: Monitor*

Amitriptylin oxide: CNS Depressants may increase CNS depressant effects of Amitriptylin oxide. *Risk C: Monitor*

Amoxapine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Antihypaciviral Combination Products: May increase serum concentrations of Midazolam. Management: Oral midazolam is contraindicated with antihypaciviral combination products. Avoid use with

nasal midazolam. Consider alternatives to use with other routes of midazolam (IV, IM) when possible. Consider use of lower midazolam doses if combined. *Risk X: Avoid*

ARIPiprazole Lauroxil: May increase CNS depressant effects of Benzodiazepines. ARIPiprazole Lauroxil may increase hypotensive effects of Benzodiazepines. Specifically, the risk of orthostatic hypotension may be increased. *Risk C: Monitor*

ARIPiprazole: May increase CNS depressant effects of Benzodiazepines. ARIPiprazole may increase hypotensive effects of Benzodiazepines. Specifically, orthostatic hypotension may be increased. *Risk C: Monitor*

Asenapine: CNS Depressants may increase CNS depressant effects of Asenapine. *Risk C: Monitor*

Atorvastatin: May increase serum concentrations of Midazolam. *Risk C: Monitor*

Azelastine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Baclofen: CNS Depressants may increase CNS depressant effects of Baclofen. *Risk C: Monitor*

Barbiturates: CNS Depressants may increase CNS depressant effects of Barbiturates. *Risk C: Monitor*

Belumosudil: May increase serum concentrations of CYP3A4 Substrates (Narrow Therapeutic Index/Sensitive with Inhibitors). Management: Avoid coadministration of belumosudil with these narrow therapeutic index/sensitive with inhibitors substrates of CYP3A4. If coadministration cannot be avoided, refer to substrate labeling and adjust the dose as instructed. *Risk D: Consider Therapy Modification*

Benperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Benzodiazepines: CNS Depressants may increase CNS depressant effects of Benzodiazepines. *Risk C: Monitor*

Beta-Acetyldigoxin: Benzodiazepines may increase serum concentrations of Beta-Acetyldigoxin. *Risk C: Monitor*

Blonanserin: CNS Depressants may increase CNS depressant effects of Blonanserin. Management: Use caution if coadministering blonanserin and CNS depressants; dose reduction of the other CNS depressant may be required. Strong CNS depressants should not be coadministered with blonanserin. *Risk D: Consider Therapy Modification*

Brexanolone: CNS Depressants may increase CNS depressant effects of Brexanolone. *Risk C: Monitor*

Brimonidine (Ophthalmic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brimonidine (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brivaracetam: CNS Depressants may increase CNS depressant effects of Brivaracetam. *Risk C: Monitor*

Bromopride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bromperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brompheniramine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Buclicine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bulevirtide: May increase serum concentrations of CYP3A4 Substrates (Narrow Therapeutic Index/Sensitive with Inhibitors). *Risk C: Monitor*

Buprenorphine: CNS Depressants may increase CNS depressant effects of Buprenorphine. Management: Avoid coadministration of buprenorphine and other CNS depressants if possible. If combined, consider adjustments to induction, increased monitoring, and co-prescription of an opioid overdose reversal agent. *Risk D: Consider Therapy Modification*

BusPIRone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Cannabinoid-Containing Products: CNS Depressants may increase CNS depressant effects of Cannabinoid-Containing Products. *Risk C: Monitor*

CarBAMazepine: CNS depressant effects of CarBAMazepine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Carbinoxamine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Carisoprodol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Cenobamate: CNS depressant effects of Cenobamate and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Certoparin: May increase serum concentrations of Benzodiazepines. *Risk C: Monitor*

Cetirizine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk D: Consider Therapy Modification*

Chloral Hydrate-Chloral Betaine: CNS Depressants may increase CNS depressant effects of Chloral Hydrate-Chloral Betaine. Management: Consider alternatives to the use of chloral hydrate or chloral betaine and additional CNS depressants. If combined, consider a dose reduction of either agent and monitor closely for enhanced CNS depressive effects. *Risk D: Consider Therapy Modification*

Chlormethiazole: May increase CNS depressant effects of CNS Depressants. Management: Monitor closely for evidence of excessive CNS depression. The chlormethiazole labeling states that an appropriately reduced dose should be used if such a combination must be used. *Risk D: Consider Therapy Modification*

Chlorphenesin Carbamate: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Chlorpheniramine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

ChlorproMAZINE: CNS depressant effects of ChlorproMAZINE and CNS Depressants may increase with coadministration. Management: If possible, discontinue the use of other CNS depressants prior to initiating chlorpromazine; other CNS depressants can be re-started at low doses and titrated slowly as needed. Monitor for CNS depression (eg, sedation, somnolence) with any combined use. *Risk D: Consider Therapy Modification*

Chlorprothixene: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Chlorzoxazone: CNS depressant effects of Chlorzoxazone and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Cinnarizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Clemastine: CNS depressant effects of Clemastine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Clofazimine: May increase serum concentrations of CYP3A4 Substrates (High risk with Inhibitors). *Risk C: Monitor*

ClomiPRAMINE: CNS depressant effects of ClomiPRAMINE and CNS Depressants may increase with coadministration. *Risk C: Monitor*

CloNIDine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Clothiapine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

CloZAPine: Benzodiazepines may increase adverse/toxic effects of CloZAPine. *Risk C: Monitor*

CloZAPine: CNS Depressants may increase CNS depressant effects of CloZAPine. *Risk C: Monitor*

CNS Depressants: May increase CNS depressant effects of Benzodiazepines. *Risk C: Monitor*

Cobicistat: May increase serum concentrations of Midazolam. Management: Oral midazolam is contraindicated with cobicistat. Avoid use with nasal midazolam. Consider alternatives to use with other

routes of midazolam (IV, IM) when possible. Consider use of lower midazolam doses if combined. *Risk X: Avoid*

Corticosteroids (Orally Inhaled): Benzodiazepines may increase adverse/toxic effects of Corticosteroids (Orally Inhaled). Specifically, the risk of pneumonia may be increased. *Risk C: Monitor*

Cyclizine: CNS depressant effects of Cyclizine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Cyclobenzaprine: CNS depressant effects of CNS Depressants and Cyclobenzaprine may increase with coadministration. *Risk C: Monitor*

CYP3A4 Inducers (Moderate): May decrease serum concentrations of Midazolam. *Risk C: Monitor*

CYP3A4 Inducers (Strong): May decrease serum concentrations of Midazolam. *Risk C: Monitor*

CYP3A4 Inhibitors (Moderate): May increase serum concentrations of Midazolam. Management: Avoid concomitant use of nasal midazolam and moderate CYP3A4 inhibitors. Consider alternatives to use with oral midazolam whenever possible and consider using lower midazolam doses. Monitor patients for sedation and respiratory depression if combined. *Risk D: Consider Therapy Modification*

CYP3A4 Inhibitors (Strong): May increase serum concentrations of Midazolam. Management: Avoid use of nasal midazolam and strong CYP3A4 inhibitors whenever possible, and consider alternatives to use with other routes of midazolam (oral, IV, IM). If combined, consider lower midazolam doses and monitor for increased midazolam toxicities. *Risk D: Consider Therapy Modification*

CYP3A4 Inhibitors (Weak): May increase serum concentrations of Midazolam. *Risk C: Monitor*

Cyproheptadine: CNS depressant effects of Cyproheptadine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Dantrolene: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Daridorexant: May increase CNS depressant effects of CNS Depressants. Management: Dose reduction of daridorexant and/or any other CNS depressant may be necessary. Use of daridorexant with alcohol is not recommended, and the use of daridorexant with any other drug to treat insomnia is not recommended. *Risk D: Consider Therapy Modification*

Desipramine: CNS Depressants may increase CNS depressant effects of Desipramine. *Risk C: Monitor*

Deutetrabenazine: CNS depressant effects of CNS Depressants and Deutetrabenazine may increase with coadministration. *Risk C: Monitor*

Dexbrompheniramine: CNS Depressants may increase CNS depressant effects of Dexbrompheniramine. *Risk C: Monitor*

Dexchlorpheniramine: CNS depressant effects of CNS Depressants and Dexchlorpheniramine may increase with coadministration. *Risk C: Monitor*

Dexmedetomidine: CNS Depressants may increase CNS depressant effects of Dexmedetomidine. Management: Monitor for increased CNS depression during coadministration of dexmedetomidine and CNS depressants, and consider dose reductions of either agent to avoid excessive CNS depression. *Risk D: Consider Therapy Modification*

Difelikefalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Difenoxin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Dihydralazine: CNS Depressants may increase hypotensive effects of Dihydralazine. *Risk C: Monitor*

Dimenhydrinate: CNS depressant effects of CNS Depressants and Dimenhydrinate may increase with coadministration. *Risk C: Monitor*

Dimethindene (Systemic): CNS Depressants may increase CNS depressant effects of Dimethindene (Systemic). *Risk C: Monitor*

Dimethindene (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Dinutuximab Beta: May increase serum concentrations of CYP Substrates (Narrow Therapeutic Index/Sensitive with Inhibitors). *Risk C: Monitor*

DiphenhydrAMINE (Systemic): CNS depressant effects of CNS Depressants and DiphenhydrAMINE (Systemic) may increase with coadministration. *Risk C: Monitor*

Diphenoxylate: May increase CNS depressant effects of CNS Depressants. Management: Avoid coadministration of diphenoxylate/atropine with other CNS depressants if possible. If coadministration cannot be avoided, monitor patients closely for CNS depression (eg, sedation, somnolence) with any combined use. *Risk D: Consider Therapy Modification*

Dothiepin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Doxepin (Systemic): CNS depressant effects of CNS Depressants and Doxepin (Systemic) may increase with coadministration. CNS Depressants may increase adverse/toxic effects of Doxepin (Systemic). Specifically, the risk of abnormal thinking or performing complex behaviors while asleep may be increased. Management: Monitor patients for CNS depression (eg, sedation, somnolence) if doxepin is coadministered with CNS depressants. If patients experience abnormal thinking or perform complex behaviors while asleep (eg, driving) strongly consider discontinuing doxepin. *Risk C: Monitor*

Doxepin (Topical): CNS Depressants may increase CNS depressant effects of Doxepin (Topical). *Risk C: Monitor*

Doxylamine: CNS Depressants may increase CNS depressant effects of Doxylamine. *Risk C: Monitor*

DroPERidol: May increase CNS depressant effects of CNS Depressants. Management: Consider dose reductions of droperidol or of other CNS agents (eg, opioids, barbiturates) with concomitant use. *Risk D: Consider Therapy Modification*

Elobixibat: May decrease serum concentrations of Midazolam. *Risk C: Monitor*

Elranatamab: May increase serum concentrations of CYP Substrates (Narrow Therapeutic Index/Sensitive with Inhibitors). *Risk C: Monitor*

Emedastine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk C: Monitor*

Entacapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Epcoritamab: May increase serum concentrations of CYP Substrates (Narrow Therapeutic Index/Sensitive with Inhibitors). *Risk C: Monitor*

Esketamine (Injection): CNS Depressants may increase therapeutic effects of Esketamine (Injection). Specifically, duration of esketamine injection may be prolonged. CNS Depressants may decrease adverse/toxic effects of Esketamine (Injection). *Risk C: Monitor*

Esketamine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Eszopiclone: CNS depressant effects of CNS Depressants and Eszopiclone may increase with coadministration. CNS Depressants may increase adverse/toxic effects of Eszopiclone. Specifically, the risk of daytime psychomotor impairment may be increased. Management: Consider reducing the dose of eszopiclone or other CNS depressants if these agents are used concomitantly, and monitor for excessive daytime psychomotor impairment with any combined use. Use of other sedative-hypnotics during the night is not recommended. *Risk D: Consider Therapy Modification*

Fenfluramine: CNS Depressants may increase CNS depressant effects of Fenfluramine. *Risk C: Monitor*

Flibanserin: CNS Depressants may increase CNS depressant effects of Flibanserin. *Risk C: Monitor*

Flunarizine: CNS Depressants may increase CNS depressant effects of Flunarizine. *Risk X: Avoid*

Flupentixol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

FluPHENAZine: CNS Depressants may increase CNS depressant effects of FluPHENAZine. *Risk C: Monitor*

Fusidic Acid (Systemic): May increase serum concentrations of CYP3A4 Substrates (High risk with Inhibitors). Management: Consider avoiding this combination if possible. If required, monitor patients closely for increased adverse effects of the CYP3A4 substrate. *Risk D: Consider Therapy Modification*

Gabapentin (Immediate Release): CNS Depressants may increase CNS depressant effects of Gabapentin (Immediate Release). *Risk C: Monitor*

Gabapentin Enacarbil (Extended Release): CNS Depressants may increase CNS depressant effects of Gabapentin Enacarbil (Extended Release). *Risk C: Monitor*

Ganaxolone: CNS Depressants may increase CNS depressant effects of Ganaxolone. *Risk C: Monitor*

Glofitamab: May increase serum concentrations of CYP Substrates (Narrow Therapeutic Index/Sensitive with Inhibitors). *Risk C: Monitor*

Grapefruit Juice: May increase serum concentrations of Midazolam. Management: Advise patients taking oral midazolam to avoid consumption of grapefruit juice. In patients who continue to consume grapefruit juice, monitor patients closely for increased and prolonged midazolam effects and toxicities. *Risk D: Consider Therapy Modification*

GuanFACINE: CNS Depressants may increase CNS depressant effects of GuanFACINE. *Risk C: Monitor*

Haloperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

HydroXYzine: May increase CNS depressant effects of CNS Depressants. Management: Reduce the dose of hydroxyzine or other CNS depressants if these agents are combined due to the potential for increased CNS depressant effects. *Risk D: Consider Therapy Modification*

Ilaprazole: May increase serum concentrations of Benzodiazepines. *Risk C: Monitor*

Iloperidone: CNS Depressants may increase CNS depressant effects of Iloperidone. *Risk C: Monitor*

Imipramine: CNS Depressants may increase CNS depressant effects of Imipramine. *Risk C: Monitor*

Inhalational Anesthetics: CNS Depressants may increase CNS depressant effects of Inhalational Anesthetics. *Risk C: Monitor*

Isocarboxazid: CNS Depressants may increase adverse/toxic effects of Isocarboxazid. *Risk X: Avoid*

Itraconazole: May increase serum concentrations of Midazolam. Management: Oral midazolam is contraindicated with, and for 2 weeks after, itraconazole. Avoid use with nasal midazolam. Consider alternatives to use with other routes of midazolam (IV, IM) when possible. Consider use of lower midazolam doses if combined. *Risk X: Avoid*

Ixabepilone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kava Kava: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Ketamine: CNS Depressants may increase CNS depressant effects of Ketamine. *Risk C: Monitor*

Ketoconazole (Systemic): May increase serum concentrations of Midazolam. Management: Oral midazolam is contraindicated with ketoconazole. Avoid use with nasal midazolam. Consider alternatives to use with other routes of midazolam (IV, IM) when possible. Consider use of lower midazolam doses if combined. *Risk X: Avoid*

Ketotifen (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kratom: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Lacosamide: CNS Depressants may increase CNS depressant effects of Lacosamide. *Risk C: Monitor*

Lasmiditan: CNS Depressants may increase CNS depressant effects of Lasmiditan. *Risk C: Monitor*

Lemborexant: May increase CNS depressant effects of CNS Depressants. Management: Dosage adjustments of lemborexant and of concomitant CNS depressants may be necessary when administered together because of potentially additive CNS depressant effects. Close monitoring for CNS depressant effects is necessary. *Risk D: Consider Therapy Modification*

LevETIRAcetam: CNS Depressants may increase CNS depressant effects of LevETIRAcetam. *Risk C: Monitor*

Levocetirizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Linvoseltamab: May increase serum concentrations of CYP Substrates (Narrow Therapeutic Index/Sensitive with Inhibitors). *Risk C: Monitor*

Lisuride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofepamine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofexidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lonafarnib: May increase serum concentrations of Midazolam. Management: Combined use is contraindicated. For patients who must receive midazolam, discontinue lonafarnib for 10 to 14 days before and for 2 days after midazolam administration. *Risk X: Avoid*

Loxapine: CNS Depressants may increase CNS depressant effects of Loxapine. Management: Consider reducing the dose of CNS depressants administered concomitantly with loxapine due to an increased risk of respiratory depression, sedation, hypotension, and syncope. *Risk D: Consider Therapy Modification*

Lumateperone: CNS Depressants may increase CNS depressant effects of Lumateperone. *Risk C: Monitor*

Lurasidone: CNS Depressants may increase CNS depressant effects of Lurasidone. *Risk C: Monitor*

Magnesium Sulfate: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Meclizine: CNS depressant effects of CNS Depressants and Meclizine may increase with coadministration. *Risk C: Monitor*

Melatonin: May increase sedative effects of Benzodiazepines. *Risk C: Monitor*

Melitracen: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Meprobamate: CNS depressant effects of CNS Depressants and Meprobamate may increase with coadministration. *Risk C: Monitor*

Mequitazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Metaxalone: CNS depressant effects of CNS Depressants and Metaxalone may increase with coadministration. *Risk C: Monitor*

Metergoline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methadone: Benzodiazepines may increase CNS depressant effects of Methadone. Management: Clinicians should generally avoid concurrent use of methadone and benzodiazepines when possible; any combined use should be undertaken with extra caution. *Risk D: Consider Therapy Modification*

Methocarbamol: CNS Depressants may increase CNS depressant effects of Methocarbamol. *Risk C: Monitor*

Methotrimeprazine: CNS Depressants may increase CNS depressant effects of Methotrimeprazine. Methotrimeprazine may increase CNS depressant effects of CNS Depressants. Management: Reduce the usual dose of CNS depressants by 50% if starting methotrimeprazine until the dose of methotrimeprazine is stable. Monitor patient closely for evidence of CNS depression. *Risk D: Consider Therapy Modification*

Methoxyflurane: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methsuximide: CNS Depressants may increase CNS depressant effects of Methsuximide. *Risk C: Monitor*

Metoclopramide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

MetyroSINE: CNS Depressants may increase sedative effects of MetyroSINE. *Risk C: Monitor*

Mianserin: CNS Depressants may increase CNS depressant effects of Mianserin. *Risk C: Monitor*

Milsaperidone: CNS Depressants may increase CNS depressant effects of Milsaperidone. *Risk C: Monitor*

Minocycline (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Mirogabalin: CNS Depressants may increase CNS depressant effects of Mirogabalin. *Risk C: Monitor*

Mirtazapine: Benzodiazepines may increase CNS depressant effects of Mirtazapine. *Risk X: Avoid*

Molindone: CNS Depressants may increase CNS depressant effects of Molindone. *Risk C: Monitor*

Mosunetuzumab: May increase serum concentrations of CYP Substrates (Narrow Therapeutic Index/Sensitive with Inhibitors). *Risk C: Monitor*

Moxonidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nabilone: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Nalfurafine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nefopam: CNS Depressants may increase CNS depressant effects of Nefopam. *Risk C: Monitor*

Nitrous Oxide: CNS Depressants may increase CNS depressant effects of Nitrous Oxide. *Risk C: Monitor*

Nortriptyline: CNS Depressants may increase CNS depressant effects of Nortriptyline. *Risk C: Monitor*

Noscapine: CNS Depressants may increase adverse/toxic effects of Noscapine. *Risk X: Avoid*

OLANZapine: Benzodiazepines may increase adverse/toxic effects of OLANZapine. Management: Monitor closely for hypotension, respiratory or central nervous system depression, and bradycardia if olanzapine is combined with benzodiazepines. Use of parenteral benzodiazepines with IM olanzapine is not recommended. *Risk C: Monitor*

OLANZapine: CNS depressant effects of OLANZapine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Olopatadine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Opicapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Opioid Agonists: CNS Depressants may increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and benzodiazepines or other CNS depressants when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Opipramol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Orphenadrine: CNS Depressants may increase CNS depressant effects of Orphenadrine. *Risk C: Monitor*

Oxatomide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Oxomemazine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Oxybate (Mixed Salts and Sodium Salt): Benzodiazepines may increase CNS depressant effects of Oxybate (Mixed Salts and Sodium Salt). *Risk X: Avoid*

Paliperidone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Paraldehyde: CNS Depressants may increase CNS depressant effects of Paraldehyde. *Risk X: Avoid*

Perampanel: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Periciazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Perphenazine: CNS depressant effects of CNS Depressants and Perphenazine may increase with coadministration. *Risk C: Monitor*

Pheniramine: CNS Depressants may increase CNS depressant effects of Pheniramine. *Risk C: Monitor*

Phenyltoloxamine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pholcodine: CNS Depressants may increase CNS depressant effects of Pholcodine. *Risk C: Monitor*

Pimozide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pipamperone: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Piribedil: CNS Depressants may increase CNS depressant effects of Piribedil. *Risk C: Monitor*

Pizotifen: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pomalidomide: CNS Depressants may increase CNS depressant effects of Pomalidomide. *Risk C: Monitor*

Pramipexole: CNS Depressants may increase sedative effects of Pramipexole. *Risk C: Monitor*

Pregabalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Procarbazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Prochlorperazine: CNS Depressants may increase CNS depressant effects of Prochlorperazine. *Risk C: Monitor*

Promazine: CNS Depressants may increase CNS depressant effects of Promazine. *Risk C: Monitor*

Promethazine: CNS Depressants may increase CNS depressant effects of Promethazine. Management: Discontinue or reduce the dose of other CNS depressants during promethazine therapy. *Risk D: Consider Therapy Modification*

Propofol: CNS Depressants may increase CNS depressant effects of Propofol. *Risk C: Monitor*

Protease Inhibitors: May increase serum concentrations of Midazolam. Management: Oral midazolam is contraindicated with protease inhibitors. Avoid use with nasal midazolam. Consider alternatives to use with other routes of midazolam (IV, IM) when possible. Consider use of lower midazolam doses if combined. *Risk X: Avoid*

Protriptyline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pyrilamine (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

QUetiapine: CNS Depressants may increase CNS depressant effects of QUetiapine. *Risk C: Monitor*

Ramelteon: CNS Depressants may increase CNS depressant effects of Ramelteon. *Risk C: Monitor*

Ranitidine: May increase serum concentrations of Midazolam. *Risk C: Monitor*

Reserpine: CNS Depressants may increase CNS depressant effects of Reserpine. *Risk C: Monitor*

Rilmenidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Risperidone: CNS depressant effects of CNS Depressants and Risperidone may increase with coadministration. *Risk C: Monitor*

Ritlecitinib: May increase serum concentrations of CYP3A4 Substrates (Narrow Therapeutic Index/Sensitive with Inhibitors). *Risk C: Monitor*

Ropeginterferon Alfa-2b: CNS Depressants may increase adverse/toxic effects of Ropeginterferon Alfa-2b. Specifically, the risk of neuropsychiatric adverse effects may be increased. Management: Avoid coadministration of ropeginterferon alfa-2b and other CNS depressants. If this combination

cannot be avoided, monitor patients for neuropsychiatric adverse effects (eg, depression, suicidal ideation, aggression, mania). *Risk D: Consider Therapy Modification*

ROPINIrole: CNS Depressants may increase sedative effects of ROPINIrole. *Risk C: Monitor*

Rotigotine: CNS Depressants may increase sedative effects of Rotigotine. *Risk C: Monitor*

Scopolamine: CNS Depressants may increase CNS depressant effects of Scopolamine. Management: Avoid coadministration of scopolamine with other CNS depressants if possible. If coadministration is required, monitor patients for excessive CNS depression. *Risk D: Consider Therapy Modification*

Spironolactone: May increase serum concentrations of CYP3A4 Substrates (Narrow Therapeutic Index/Sensitive with Inhibitors). *Risk C: Monitor*

Stiripentol: CNS Depressants may increase CNS depressant effects of Stiripentol. *Risk C: Monitor*

Suvorexant: CNS Depressants may increase CNS depressant effects of Suvorexant. *Risk C: Monitor*

Talquetamab: May increase serum concentrations of CYP Substrates (Narrow Therapeutic Index/Sensitive with Inhibitors). *Risk C: Monitor*

Tarlatamab: May increase serum concentrations of CYP Substrates (Narrow Therapeutic Index/Sensitive with Inhibitors). *Risk C: Monitor*

Tasimelteon: CNS Depressants may increase CNS depressant effects of Tasimelteon. *Risk C: Monitor*

Teclistamab: May increase serum concentrations of CYP Substrates (Narrow Therapeutic Index/Sensitive with Inhibitors). *Risk C: Monitor*

Teduglutide: May increase serum concentrations of Benzodiazepines. *Risk C: Monitor*

Tetrabenazine: CNS Depressants may increase CNS depressant effects of Tetrabenazine. *Risk C: Monitor*

Thalidomide: CNS Depressants may increase CNS depressant effects of Thalidomide. *Risk X: Avoid*

Theophylline Derivatives: May decrease therapeutic effects of Benzodiazepines. *Risk C: Monitor*

Thioridazine: CNS Depressants may increase CNS depressant effects of Thioridazine. *Risk C: Monitor*

Thiothixene: CNS Depressants may increase CNS depressant effects of Thiothixene. *Risk C: Monitor*

TiaGABine: CNS Depressants may increase CNS depressant effects of TiaGABine. *Risk C: Monitor*

TiZANidine: CNS Depressants may increase CNS depressant effects of TiZANidine. *Risk C: Monitor*

Tolcapone: CNS Depressants may increase CNS depressant effects of Tolcapone. *Risk C: Monitor*

Topiramate: CNS Depressants may increase CNS depressant effects of Topiramate. CNS Depressants may increase adverse/toxic effects of Topiramate. Specifically, cognitive dysfunction and neuropsychiatric adverse effects may be increased. *Risk C: Monitor*

Tradipitant: CNS Depressants may increase CNS depressant effects of Tradipitant. *Risk C: Monitor*

TraZODone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Treosulfan: May increase serum concentrations of CYP3A4 Substrates (Narrow Therapeutic Index/Sensitive with Inhibitors). *Risk C: Monitor*

Trifluoperazine: CNS Depressants may increase CNS depressant effects of Trifluoperazine. *Risk C: Monitor*

Trimeprazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Trimethobenzamide: CNS Depressants may increase CNS depressant effects of Trimethobenzamide. Management: Avoid coadministration of trimethobenzamide with other CNS depressants if possible. If coadministration cannot be avoided, monitor patients for CNS adverse effects (eg, depression, disorientation, seizures). *Risk D: Consider Therapy Modification*

Trimipramine: CNS Depressants may increase CNS depressant effects of Trimipramine. *Risk C: Monitor*

Triprolidine: CNS Depressants may increase CNS depressant effects of Triprolidine. *Risk C: Monitor*

Trofinetide: May increase serum concentrations of CYP3A4 Substrates (Narrow Therapeutic Index/Sensitive with Inhibitors). *Risk C: Monitor*

Valerian: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Valproic Acid and Derivatives: CNS Depressants may increase CNS depressant effects of Valproic Acid and Derivatives. *Risk C: Monitor*

Vigabatrin: CNS Depressants may increase CNS depressant effects of Vigabatrin. *Risk C: Monitor*

Xanomeline: May increase serum concentrations of CYP3A4 Substrates (Narrow Therapeutic Index/Sensitive with Inhibitors). *Risk C: Monitor*

Yohimbine: May decrease therapeutic effects of Antianxiety Agents. *Risk C: Monitor*

Zaleplon: CNS Depressants may increase CNS depressant effects of Zaleplon. Management: Coadministration of zaleplon with other sedative-hypnotics at bedtime or in the middle of the night is not recommended. If zaleplon is coadministered with other CNS depressants at other times, monitor for CNS depression; dose reductions may be required. *Risk D: Consider Therapy Modification*

Ziconotide: CNS Depressants may increase CNS depressant effects of Ziconotide. *Risk C: Monitor*

Ziprasidone: CNS Depressants may increase CNS depressant effects of Ziprasidone. *Risk C: Monitor*

Zolpidem: CNS Depressants may increase CNS depressant effects of Zolpidem. Management: Reduce the Intermezzo brand sublingual zolpidem adult dose to 1.75 mg for men who are also receiving other CNS depressants. No such dose change is recommended for women. Avoid use with other CNS depressants at bedtime; avoid use with alcohol. *Risk D: Consider Therapy Modification*

Zonisamide: CNS Depressants may increase CNS depressant effects of Zonisamide. *Risk C: Monitor*

Zopiclone: CNS Depressants may increase CNS depressant effects of Zopiclone. *Risk C: Monitor*

Zuclopenthixol: May increase CNS depressant effects of CNS Depressants. Management: Avoid use of high-dose hypnotics or other CNS depressants together with zuclopenthixol. *Risk D: Consider Therapy Modification*

Zuranolone: May increase CNS depressant effects of CNS Depressants. Management: Consider alternatives to the use of zuranolone with other CNS depressants or alcohol. If combined, consider a zuranolone dose reduction and monitor patients closely for increased CNS depressant effects. *Risk D: Consider Therapy Modification*

Food Interactions

Oral: Grapefruit juice may increase serum concentrations of midazolam. Management: Avoid concurrent use of grapefruit juice with oral midazolam.

Pregnancy Considerations

Midazolam crosses the placenta (Bach 1989; Kanto 1983).

In utero exposure to benzodiazepines has the potential to cause harm to the fetus. Teratogenic effects have been observed in some studies; however, a clear association has not been reported and additional data are needed (Bellantuono 2013; Chuang 2024; Freeman 2018; Grigoriadis 2019; Tinker 2019; Wu 2024). Exposure to a benzodiazepine late in pregnancy may cause neonatal sedation (hypotonia, lethargy, respiratory depression) and/or symptoms of neonatal withdrawal (feeding difficulties, hyperreflexia, inconsolable crying, irritability, restlessness, tremors). Monitor newborns exposed to midazolam in utero for adverse events. Data related to long-term effects on neurodevelopment following maternal use of benzodiazepines are inconclusive (Andrade 2024; Radojčić 2017; Sundbakk 2024; Wang 2022).

Based on animal data, repeated or prolonged use of general anesthetic and sedation medications that block N-methyl-D-aspartate (NMDA) receptors and/or potentiate gamma-aminobutyric acid (GABA) activity may affect brain development. Evaluate benefits and potential risks of fetal exposure to midazolam when duration of surgery is expected to be >3 hours (Olutoye 2018).

Due to pregnancy-induced physiologic changes, some pharmacokinetic properties of midazolam may be altered (Hebert 2008; Kanto 1984; Wilson 1987).

Midazolam use in obstetric anesthesia has been described (Hung 2022; Neuman 2013; Senel 2014; Shergill 2012). The ACOG recommends that pregnant patients should not be denied medically necessary surgery regardless of trimester. If the procedure is elective, it should be delayed until after delivery (ACOG 2019).

Treatment of status epilepticus during pregnancy should consider gestational age and etiology of seizures (eg, eclampsia versus other causes) (NCS [Brophy 2012]; Rajiv 2019).

Data related to the treatment of status epilepticus in pregnancy are limited; use of midazolam may be considered (NCS [Brophy 2012]; Rajiv 2019).

Epilepsy is associated with adverse maternal and fetal outcomes (Kuang 2024; Mazzone 2023). Convulsive seizures should be minimized to reduce risks to the fetus and pregnant patient. Use caution if removing or replacing an effective seizure medication in patients who become pregnant during therapy (Pack 2024).

Data collection to monitor pregnancy outcomes following exposure to antiepileptic drugs is ongoing. Encourage patients to enroll in the North American Antiepileptic Drug (NAAED) Pregnancy Registry (1-888- 233-2334 or www.aedpregnancyregistry.org.)

Breastfeeding Considerations

Midazolam and hydroxymidazolam are present in breast milk.

Data related to the presence of midazolam in breast milk are available following administration to 12 postpartum women for sleep. Breast milk and maternal serum were sampled in the morning, ~7 hours after the maternal dose. Following accidental administration of 2 oral doses (total dose: 30 mg) to 1 woman, breast milk concentrations of midazolam were 0.0001 mg/mL (30 nmol/L). Midazolam was not present in the breast milk of the 11 other patients (Matheson 1990).

- Using a milk concentration of 0.0001 mg/mL (30 nmol/L), the estimated exposure to the breastfed infant would be 0.0015 mg/kg/day (relative infant dose [RID] 0.35% based on a weight-adjusted maternal dose of 30 mg).

- In general, breastfeeding is considered acceptable when an RID of a medication is <10% (Anderson 2016; Ito 2000).

- In most reports, midazolam and hydroxymidazolam concentrations in breast milk are below the limit of quantification 4 hours after a single dose (Koitabashi 1997; Matheson 1990).

CNS depression was reported in infants following exposure to benzodiazepines via breast milk (study included exposures to midazolam) (Kelly 2012); poor feeding and poor weight gain may also occur.

According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and the benefits of treatment to the mother. Infants exposed to midazolam via breast milk should be monitored for sedation, poor feeding, and poor weight gain.

When used for sedation prior to a procedure, available guidelines suggest waiting for ≥4 hours after a maternal dose of midazolam to continue breastfeeding (Shergill 2012); some manufacturers suggest patients express and discard milk for 4 to 8 hours after midazolam administration to decrease potential exposure to a breastfed infant. The Academy of Breastfeeding Medicine recommends postponing elective surgery until milk supply and breastfeeding are established. Milk should be expressed ahead of surgery when possible. In general, when the child is healthy and full term, breastfeeding may resume, or milk may be expressed once the mother is awake and in recovery. For children who are at risk for apnea, hypotension, or hypotonia, milk may be saved for later use when the child is at lower

risk (ABM [Reece-Stremtan 2017]). Small supplemental doses of midazolam used during cesarean delivery should not prevent breastfeeding once the mother is stable and alert (ABM [Martin 2018]).

Dietary Considerations

Avoid grapefruit juice with oral syrup.

Monitoring Parameters

Level of sedation, respiratory rate, heart rate, blood pressure, oxygen saturation (ie, pulse oximetry).

Critically ill patients: Assess and adjust sedation according to scoring system (eg, Richmond Agitation-Sedation Scale [RASS] or Sedation-Agitation Scale [SAS]) or clinical effect (eg, ventilator synchrony) (SCCM [Devlin 2018]).

Mechanism of Action

Short-acting benzodiazepine (Griffin 2013). Binds to stereospecific benzodiazepine receptors on the postsynaptic GABA neuron at several sites within the central nervous system, including the limbic system and reticular formation. Enhancement of the inhibitory effect of GABA on neuronal excitability results by increased neuronal membrane permeability to chloride ions. This shift in chloride ions results in hyperpolarization (a less excitable state) and stabilization. Benzodiazepine receptors and effects appear to be linked to the GABA-A receptors. Benzodiazepines do not bind to GABA-B receptors (Brunton 2011).

Pharmacokinetics (Adult Data Unless Noted)

Onset of action:

Sedation/anesthesia:

IM: Children: Within 5 minutes; Adults: ~15 minutes.

IV: 1 to 5 minutes (dose dependent) (Barr 2013; Horn 2004; Roberts 2019).

Oral: 10 to 20 minutes.

Intranasal (nasal spray): Within 10 minutes.

Intranasal (solution, injection): Children: 5.55 ± 2.22 minutes (Lee-Kim 2004); Adults: Within 5 minutes.

Peak effect: IM: Children: 15 to 30 minutes; Adults: 30 to 60 minutes; IV: 3 to 5 minutes; Intranasal (nasal spray): 30 minutes; Intranasal (solution, injection): Children: 10 minutes (al-Rakaf 2001).

Duration:

Classified as a short-acting benzodiazepine; classification based on benzodiazepines with half-life of 1 to 12 hours (Griffin 2013).

Indication-specific durations:

Sedation:

Children: Intranasal (solution, injection): 23.1 minutes (Chiaretti 2011).

Adults: IM: 20 to 120 minutes; IV: 7 to 75 minutes (Bell 1991).

Absorption: IM: Rapid, complete; Intranasal (nasal spray): Rapid; Oral, Intranasal (solution, injection): Rapid (Lee-Kim 2004).

Distribution: Widely distributed in body including CSF; V_d :

Preterm infants (n=24; GA: 26 to 34 weeks; PNA: 3 to 11 days): IV: Median: 1.1 L/kg (range: 0.4 to 4.2 L/kg) (de Wildt 2001).

Infants and Children 6 months to 16 years: IV: 1.24 to 2.02 L/kg.

Adults: 1 to 3.1 L/kg; increased in females, elderly, and obesity.

Protein binding: ~97%, primarily albumin; in patients with cirrhosis, protein binding is reduced with a free fraction of ~5% (Trouvin 1988).

Metabolism: Extensively hepatic via CYP3A4; 60% to 70% of biotransformed midazolam is the active metabolite 1-hydroxy-midazolam (or alpha-hydroxymidazolam).

Bioavailability: Oral: 40% to 50% (Kanto 1985), ~36% (children); IM: >90%; Intranasal (nasal spray): 44%; Intranasal (solution, injection): Children: ~60%; Rectal: Children: ~40% to 65% (mean: 52%) (Clausen 1988).

Half-life elimination:

Preterm infants (n=24; GA: 26 to 34 weeks; PNA: 3 to 11 days): IV: Median: 6.3 hours (range: 2.6 to 17.7 hours) (de Wildt 2001).

Neonates: 4 to 12 hours; seriously ill neonates: 6.5 to 12 hours.

Children: IV: 2.9 to 4.5 hours; Syrup: 2.2 to 6.8 hours.

Adults: 3 hours (range: 1.8 to 6.4 hours); IM: 4.2 ± 1.87 hours; Intranasal (nasal spray): 2.1 to 6.2 hours.

Time to peak, serum: IM: ~0.5 hours; Intranasal (nasal spray): Median 17.3 minutes (7.8 to 28.2 minutes); Oral: 0.17 to 2.65 hours.

Excretion: Intranasal (nasal spray): Urine (primarily as glucuronide conjugates of the hydroxylated metabolites); IV, IM: Urine (primarily as metabolites); Oral: Urine (~90% within 24 hours; primarily [60% to 70%] as glucuronide conjugates of the hydroxylated metabolites; <0.03% as unchanged drug); feces (~2% to 10% over 5 days) (Kanto 1985; Smith 1981).

Clearance:

Preterm infants (n=24; GA: 26 to 34 weeks; PNA: 3 to 11 days): Median: 1.8 mL/minute/kg (range: 0.7 to 6.7 mL/minute/kg) (de Wildt 2001).

Neonates <39 weeks GA: 1.17 mL/minute/kg.

Neonates >39 weeks GA: 1.84 mL/minute/kg.

Seriously ill neonates: 1.2 to 2 mL/minute/kg.

Infants >3 months of age: 9.1 mL/minute/kg.

Children >1 year of age: 3.2 to 13.3 mL/minute/kg.

Healthy adults: 4.2 to 9 mL/minute/kg.

Adults with acute renal failure: 1.9 mL/minute/kg.

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Altered kidney function: Elimination half-life is prolonged and clearance is reduced. In patients with renal failure, reduced elimination of active hydroxylated metabolites leads to drug accumulation and prolonged sedation.

Hepatic function impairment: In patients with cirrhosis, following oral administration, C_{max} and bioavailability were 43% and 100% higher, and following IV administration clearance was reduced 50%, half-life increased 2.5-fold, and V_d increased by 20%.

Older adult: IM, IV (mean age: 73 years): Plasma half-life was ~2-fold higher; mean V_d increased consistently between 15% to 100%, and mean clearance decreased ~25%; Intranasal (nasal spray) (>65 years of age): AUC and C_{max} increased 21% to 45%.

Heart failure: Following oral administration (7.5 mg), half-life increased 43%. Two-fold increase in the elimination half-life, a 25% decrease in the plasma clearance and a 40% increase in V_d following parenteral administration.

Obesity: Mean half-life is prolonged (5.9 hours) and V_d increased ~50%.

Therapeutic hypothermia: Hypothermia may be associated with a 5-fold increase in plasma concentrations, likely due to decreased total body clearance (~11% decline in clearance for every centigrade decrease in body temperature below 36.5°C [97.7°F]) (Fukuoka 2004; Hostler 2010). During the rewarming phase, plasma concentrations decrease (~11% decline in plasma concentrations as temperature increases from 33°C to 37°C [91.4°F to 98.6°F]) (Bjelland 2014). The V_d in hypothermia (core body temperature <35°C [<95°F]) may decrease ~45% during rewarming phase (Fukuoka 2004); however, some reports found no significant impact of hypothermia on total body clearance (Bastiaans 2013) or V_d (Hostler 2010).

Medication Guide and/or Vaccine Information Statement (VIS)

An FDA-approved patient medication guide, which is available with the product and as follows, must be dispensed with this medication:

Nayzilam: https://www.accessdata.fda.gov/drugsatfda_docs/label/2023/211321s008lbl.pdf#page=26

Patient Counseling Points

(For additional information see "Midazolam: Patient drug information")

What is this drug used for?

- It is used to calm you before a procedure.
- It is used to cause sleep during a procedure.
- It is used to treat seizures.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

Nose spray:

- Feeling sleepy
- Nose or throat irritation
- Runny nose
- Headache
- Memory problems or loss

Liquid (syrup) and injection:

- Feeling sleepy
- Feeling tired or weak
- Memory problems or loss

Liquid (syrup):

- Upset stomach or throwing up

Injection:

- Irritation where this drug is given

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- Chest pain or pressure or slow, shallow, or trouble breathing
- Shakiness
- Twitching

- Severe dizziness or passing out
- Feeling agitated
- Trouble controlling body movements
- Depression; feeling nervous, restless, or grouchy; panic attacks; other changes in mood or behavior; suicidal thoughts or actions
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Dormicum;
- (AR) Argentina: Dormicum | Dormonid | Midalan | Ormir | Rem chobet;
- (AT) Austria: Dormicum | Midazolam mayrhofer;
- (AU) Australia: Hypnovel;
- (BD) Bangladesh: Anquil | Benquil | Dormax | Dormicum | Dormitol | Hypnocum | Hypnofast | Midolam | Midzo | Milam | Mizolam;
- (BE) Belgium: Dormicum;
- (BG) Bulgaria: Dormicum;
- (BR) Brazil: Dormire | Dormium | Dormonid | Maleato de Midazolam | Midazolam | Prontomid;
- (CH) Switzerland: Dormicum;
- (CL) Chile: Dormonid | Noctura | Terap;
- (CN) China: Dormicum | Li yue xi;
- (CO) Colombia: Dormicum | Dormipron | Espatabs | Midazolam | Mizolam;
- (CR) Costa Rica: Dormicum;
- (CZ) Czech Republic: Dormicum | Midazolam accord | Midazolam avmc;
- (DE) Germany: Dormicum | Midazolam accord | Midazolam delta select;
- (DO) Dominican Republic: Dormicum | Dormire;
- (EC) Ecuador: Dormicum;
- (EE) Estonia: Dormicum;
- (EG) Egypt: Dormicum | Midathetic;
- (ES) Spain: Dormicum;
- (FI) Finland: Dormicum | Midazolam accord | Midazolam alparma;
- (GB) United Kingdom: Midazolam;
- (HK) Hong Kong: Dormicum | Midazolam Rotexmedica;
- (HR) Croatia: Dormicum;

- (HU) Hungary: Dormicum | Midanxil;
- (IL) Israel: Midolam;
- (IN) India: Mezolam;
- (JO) Jordan: Dormicum;
- (JP) Japan: Midafresa;
- (KE) Kenya: Dormicum;
- (KR) Korea, Republic of: Dormicum;
- (KW) Kuwait: Dormicum;
- (LB) Lebanon: Midazolam;
- (LT) Lithuania: Dormicum;
- (LU) Luxembourg: Dormicum;
- (LV) Latvia: Dormicum;
- (MX) Mexico: Dormicum | Midazolam;
- (MY) Malaysia: Domi | Dormicum | Fulsed | Midazolam;
- (NL) Netherlands: Dormicum | Senozam;
- (NO) Norway: Dormicum;
- (NZ) New Zealand: Hypnovel | Midazolam;
- (PA) Panama: Dormicum;
- (PE) Peru: Amadaz | Dormonid | Noctura;
- (PH) Philippines: Dormicum;
- (PK) Pakistan: Cazolam | Dormicum | Hyrest | Milam;
- (PL) Poland: Dormicum | Midazolam | Midazolam accord;
- (PR) Puerto Rico: Midazolam | Midazolam HCL | Nayzilam | Versea;
- (PT) Portugal: Dormicum | Midazolam b. braun | Zolamid;
- (PY) Paraguay: Dormicum | Midazolam dutriec | Midazolam icu vita | Noctura | Sedancor;
- (QA) Qatar: Dilemy | Dormicum | Dormicum (IV/IM/Rectal) | Epistatus | Hypnovel | Sedalam (MS Pharma) | Zolamid;
- (RO) Romania: Dormicum;
- (RU) Russian Federation: Dormicum | Fulsed;
- (SE) Sweden: Midazolam accord;
- (SG) Singapore: Dormicum;
- (SI) Slovenia: Dormicum;
- (SK) Slovakia: Dormicum;
- (SR) Suriname: Midazolam;
- (TH) Thailand: Dormicum;
- (TR) Turkey: Dormicum;
- (TW) Taiwan: Dormicum | Uzolam;
- (UG) Uganda: Dormicum;
- (UY) Uruguay: Andavil | Dorminox | Hipnazolam | Midazolam | Midazolam icu vita;
- (VE) Venezuela, Bolivarian Republic of: Doricum;
- (ZA) South Africa: Accord midazolam | Dormicum | Micro midazolam;
- (ZW) Zimbabwe: Dormicum

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